Bladder Cancer & Urinary Diversion Rapid Revision

Campbell-Walsh-Wein · Chapters 124, 129–135

For MCh Urology theory & viva · NEET-SS / competitive MCQs

How to use (total ≈ 5h 36min even at a slow, careful pace)

- Each chapter reads top-to-bottom in the order you would write an essay answer: definition → pathology → diagnosis → management → complications → follow-up. The same material serves theory and viva.
- Boxed **ALGORITHM** and shaded **KEY POINT** panels carry the reasoning examiners ask about ("why do we…?").
- **MCQ one-liners** end each chapter: re-read these the day before the paper.
- The last section puts the big treatment choices side by side for essay questions.

Conventions: **underlined bold** = number to memorise · MCQ = exam trivia only · → *Ch 132* = cross-reference · † = not in the source chapter notes (added because it is essential in practice). Abbreviations are spelled out at first use in each chapter.

Chapter	Topic	Slow-reader time
129	Tumours of the Urinary Bladder	31 min
130	Management Strategies for NMIBC	38 min
131	Muscle-Invasive & Metastatic Bladder Cancer	44 min
132	Surgical Management of Bladder Cancer	28 min
133	Intestinal Segments in Urinary Diversion	44 min
134	Continent Urinary Diversion	46 min
135	Minimally Invasive Urinary Diversion	27 min
124	Bladder Surgery for Benign Disease (incl. vesicovaginal fistula)	70 min
★	Putting it together: the big treatment choices	8 min
	Total (23624 words, at 70 words/min)	**5h 36min**

129 Tumours of the Urinary Bladder

Reading time ≈ 31 min (slow pace)

1 · EPIDEMIOLOGY (HEADLINE)

- About 550,000 new cases and 200,000 deaths worldwide (2018). **Fourth most prevalent cancer** because it is common, rarely fatal early and repeatedly recurrent → years of surveillance; **most expensive cancer per patient over a lifetime**.
- Median age at diagnosis **73**; ~9 in 10 after 55. **Male:female ≈ 3:1** – smoking, sex differences in carcinogen metabolism and sex steroids.
- **Females present later and do worse stage for stage**: haematuria is often treated as urinary tract infection (UTI), delaying referral.

2 · RISK FACTORS

The bladder is an exposure surface. Almost every risk factor is either a **chemical load** (aromatic amines → urothelial carcinoma, UC) or **chronic inflammation** (→ squamous or glandular tumours), so the exposure predicts the histology.

- **Aromatic amines** (2-naphthylamine, 4-aminobiphenyl, benzidine): hydroxylated metabolites form DNA adducts. Tobacco and occupation are the two main routes.
- **Smoking**: **30–40%** of UC; risk **2–3×**; pipes and cigars too; carcinogens found in e-cigarette users. **N-acetyltransferase 2 (NAT2) slow acetylators** are at highest risk, because NAT2 acetylation detoxifies aromatic amines. Risk falls progressively after quitting, approaching never-smokers after **>15 years**. ~20% still smoke at diagnosis; active smoking raises recurrence in non-muscle-invasive bladder cancer (NMIBC) and cancer death in muscle-invasive bladder cancer (MIBC).
- **Occupation** (5–10%): aromatic amines – dye, rubber, tobacco, leather workers, hairdressers, painters; polycyclic aromatic hydrocarbons – chimney sweeps, aluminium and petroleum workers.
- **Arsenic** in drinking water: West Bengal, Bangladesh, Taiwan.
- **Radiation**: latency 15–30 years. External beam radiation therapy (EBRT) for cervical cancer 2–4×, prostate 1.5–2×; brachytherapy smaller; intensity-modulated radiation therapy (IMRT) little or none.
- **Genetics/hereditary**: NAT2 and glutathione S-transferase mu 1 (GSTM1) deletion matter mainly with exposure; first-degree relatives 2×, no mendelian pattern. **Lynch syndrome** (mainly **MSH2**). Pathogenic germline variants in 13–24% of UC (MSH2, BRCA1/2), in high-grade not low-grade NMIBC → germline testing in selected high-grade patients.
- **Obesity**: dose-dependent risk.
- **Schistosoma haematobium** (only species linked): eggs in vesical venules → T-helper 2 inflammation, fibrosis → **squamous cell carcinoma (SCC)**; also potentiates smoking.
- **Chronic irritation**: indwelling catheter in spinal cord injury, stones, infection → SCC. **Bladder exstrophy**: malignancy 3.3–7.5%, **>90% adenocarcinoma**.
- **Drugs**: **Cyclophosphamide** – the only chemotherapeutic proven; metabolite **phosphoramide mustard** († acrolein causes haemorrhagic cystitis, not cancer); ~4.5×, dose dependent (>20 g), latency peaks at 10–14 years. **Pioglitazone** – conflicting data.
- **Diet**: fluid, coffee/tea, alcohol, red meat, vitamins, selenium – no association; supplements do not protect. Every dietary signal disappears once smoking is controlled.

3 · PATHOLOGY

Normal urothelium

4–7 cells thick: basal → intermediate → superficial **umbrella cells**, which form the urine–blood barrier.

Benign lesions (mostly reactive to chronic irritation – recognise the mimic)

Lesion	Hallmark	Malignant potential / action
Squamous/glandular metaplasia	Trigone; squamous in up to 40% of females	None; no biopsy or treatment
Inverted papilloma	Trigone; endophytic nests, no atypia; <1% of tumours	Benign, ~1% recurrence; resect
Urothelial papilloma	Delicate stalks, normal urothelium; fibroblast growth factor receptor 3 (FGFR3) mutant, TP53/RB wild type	None (formerly low-grade Ta); resect
Nephrogenic adenoma	Renal tubule-like glands after trauma, surgery, transplant, intravesical therapy; may look papillary	None; repeat resection if recurs
Leukoplakia	Keratinising squamous metaplasia; floating white flakes	**Not premalignant in bladder** (no TP53 evidence) → no follow-up
Cystitis cystica/glandularis	From proliferating von Brunn nests	Glandularis: low risk of adenocarcinoma → resect, treat cause, endoscopic surveillance

Flat (precursor) lesions

Graded by two questions: is atypia out of proportion to inflammation, and are maturation, polarity and umbrella cells preserved? **Loss of polarity = dysplasia; loss of umbrella cells = carcinoma in situ (CIS).**

Lesion	Key features	Significance
Hyperplasia (flat/papillary)	≥10 layers, no atypia	Not malignant alone; flat may precede low-grade papillary (FGFR3)
Reactive atypia	Atypia proportionate to inflammation; polarity preserved	Coexists with dysplasia/CIS → investigate
Atypia of unknown significance	Atypia disproportionate to inflammation; CK20, CD44	Limited utility
Dysplasia	Normal thickness, loss of polarity, umbrella cells kept; CK20, p53	14–19% progress; **60%** → CIS if prior cancer
CIS	Flat, high grade, noncohesive pleomorphic cells; umbrella cells lost	Precursor of invasive high-grade disease

Urothelial carcinoma: grade (2004 WHO) and the noninvasive spectrum

UC is the commonest urinary tract malignancy. NMIBC = ~**70%**: Ta (low/high grade), CIS and T1.

	papillary urothelial neoplasm of low malignant potential (PUNLMP)*	Low-grade Ta	High-grade papillary	CIS
Morphology	Papillary >7 cells, minimal atypia, solitary, trigone	Branching stalks, some atypia	Fused papillae, many mitoses	Flat, severe atypia
Genetics	TERT promoter, FGFR3	Chromosome 9 loss; FGFR3, phosphoinositide 3-kinase (PI3K), RAS	TP53/RB	RB, TP53, PTEN; **no FGFR3**
Behaviour	Recur ~20%/5 y; progress <1%	Recur 48–71%; progress 2–12%	>80% invade if untreated	With invasive tumour: 45–65% 5-y mortality

*Papillary urothelial neoplasm of low malignant potential. **Grade, not recurrence, predicts danger.**

Histological variants of UC

~80% of UC contain some mixed differentiation, most often squamous. **Altered growth patterns behave worse than their stage**; divergent differentiation mostly reflects stage.

Variant	Hallmark	Chemo-sensitive?	Management
Micropapillary (0.7–2.2%)	Tight clusters with retraction artefact; **any amount counts**; ERBB2 amplification, luminal	No	**Early radical cystectomy (RC) even for NMIBC**; transurethral resection + bacillus Calmette–Guérin (BCG) fails unless completely resected; in MIBC immediate RC
Sarcomatoid (0.3–0.6%)	Large infiltrative mass; same clone as carcinosarcoma	Poor	Immediate RC ± intraoperative radiation
Plasmacytoid (<1%; includes signet ring)	Eccentric nuclei; **intraperitoneal spread**; CDH1 truncation	Yes, but relapse common	Neoadjuvant chemotherapy (NAC) whenever possible
Nested (M:F 6:1)	Mimics von Brunn nests, inverted papilloma	No	Upstaged often; stage-matched outcome as UC
Squamous differentiation (16–22%, up to 60%)	Basal subtype	NAC active in mixed tumours	Linked to recurrence, progression
Glandular differentiation (~10%)	Gland-like spaces	–	Outcome as conventional UC

Non-urothelial malignancy

	Small cell	SCC	Adenocarcinoma	Urachal adenocarcinoma
Setting	Rare; dedifferentiated UC (not lung lineage)	2–5% in West; majority where schistosomiasis endemic	**Secondary > primary** (colon, prostate, uterus, cervix); exstrophy	~1/3 of bladder adenocarcinomas; allantoic remnant
Hallmark	Chromogranin, synaptophysin; paraneoplastic (hypercalcaemia, Cushing); TP53, RB1	Locally advanced, fewer metastases	Cribriform/glandular; **colonoscopy** to exclude primary	Dome/anterior wall; Sheldon staging
Treatment	**Cisplatin–etoposide first**, then RC or radiation	RC; preoperative radiation worsened survival	RC; no level 1 data; 5-fluorouracil–cisplatin by analogy	**En bloc dome + urachal ligament + umbilicus**

> **KEY POINT** Small cell is treated as systemic disease: primary cystectomy gives poor disease-free survival not rescued by adjuvant chemotherapy, whereas neoadjuvant chemotherapy markedly improves cancer-specific survival. SCC in spinal cord injury fell after the switch to intermittent self-catheterisation – proof that chronic inflammation drives it.

Urachal carcinoma (2016 WHO criteria): dome/anterior wall with epicentre in bladder wall; no widespread cystitis cystica/glandularis; secondary source excluded. TNM unsuitable (almost all muscle invasive by location).

Sheldon stage	Extent
I	Urachal mucosa
II	Confined to urachus
IIIA / IIIB / IIIC	Bladder / abdominal wall / peritoneum

Sheldon stage	Extent
IV	Metastatic

4 · MOLECULAR BIOLOGY

Classical two-pathway model

	Papillary pathway	Flat pathway
Origin → product	Hyperplasia → low-grade noninvasive papillary	Dysplasia → CIS → high-grade invasive
Genetics	**FGFR3** (up to 80% of low-grade Ta vs 10–20% MIBC), chromosome 9 deletion, PIK3CA (25% NMIBC)	**TP53** (up to 76% MIBC), RB1, CDKN2A, PTEN loss
Behaviour	Recurs, rarely progresses	Progresses

- HRAS/KRAS mutations are **mutually exclusive with FGFR3** (same RAS–MEK–ERK pathway) and equally frequent in NMIBC and MIBC. RB1 loss → E2F3 overexpression. PTEN loss with TP53 alteration → metastasis, poor prognosis.
- FGFR3 isoform switch IIIb → IIIc; FGFR3–TACC3 fusions are activated oncoproteins.

Expression subtypes (luminal–basal axis)

The Cancer Genome Atlas (TCGA) and others showed subtypes, resembling breast cancer, that predict behaviour and therapy independent of stage and grade.

	Luminal	Basal/squamous
Markers	FGFR3, GATA3, KRT20, PPARG, FOXA1	CK5, CK6, CK14; low GATA3, FOXA1
Behaviour	Less aggressive; papillary	Aggressive; squamous differentiation; women; T3/T4
Targets	FGFR3 inhibitors, ERBB2	Cisplatin NAC; immune checkpoint inhibitors (immune infiltrated); EGFR

- **Consensus classification (2020), six classes**: luminal papillary (24%; best, median overall survival 4 y), luminal nonspecified, luminal unstable, stroma-rich, **basal/squamous (35%, commonest)**, neuroendocrine-like (3%; RB1 loss; worst, ~1 y).
- TCGA: luminal-infiltrated responds poorly to cisplatin NAC (immunotherapy candidate); neuronal → etoposide–cisplatin; APOBEC high-mutational-load subset has better survival. MCQ Micropapillary maps to a luminal class.

5 · CLINICAL PRESENTATION AND EVALUATION

- **Painless gross haematuria in 85%**, intermittent; microscopic haematuria in nearly all. Irritative symptoms suggest CIS.
- Every episode of gross haematuria needs evaluation even if later urinalysis is negative; anticoagulants/antiplatelets do not excuse it.
- Bimanual examination at transurethral resection of bladder tumour (TURBT) if tumour appears invasive → *Ch 132*. Consider prostate-specific antigen (PSA): ~10% of recurrent gross haematuria is prostate cancer.
- After a negative work-up, risk is near zero for 6 years; repeat at 3–5 years if haematuria persists.

	Gross haematuria	Asymptomatic microhaematuria
Malignancy	~20%	2.6–5.2%
Cystoscopy	All adults	Age ≥35 years
Upper tract	Computed tomography urogram (CTU)	CTU (2012 American Urological Association, AUA); ultrasound may replace
Cytology/markers	Reasonable	Avoid routinely

Dipstick detects haem peroxidase activity: sensitive (91–100%) but specificity only ~65% → **confirm with microscopy** before evaluating.

Cystoscopy and enhanced imaging

Cystoscopy and biopsy are the gold standard; flexible office cystoscopy is as reliable as rigid. White light cystoscopy (WLC) is the standard but misses small papillary tumours and CIS.

	Blue light (HAL–BLC)	Narrow-band imaging (NBI)
Principle	Hexaminolevulinate (HAL) → porphyrins accumulate in tumour → red fluorescence under blue light (360–450 nm)	No dye; 415 and 540 nm bands absorbed by haemoglobin → vessels stand out dark
Benefit	More Ta and CIS; CIS seen only on BLC in ~¼; flexible office BLC finds recurrences missed by WLC in ~1/5	Better CIS detection; randomised trial – overall 12-month recurrence same, lower in low-risk
Status	Adjunct; only HAL is Food and Drug Administration (FDA) approved	Adjunct

Urine cytology – Paris System

Sensitivity 31–62%, specificity 94–100%: **good for high-grade, blind to low-grade**. The Paris System (2.0) standardises reporting around high-grade urothelial carcinoma (HGUC) and reduces vague "atypia" calls.

Category	Criteria
Negative for HGUC	Nuclear:cytoplasmic (N:C) ratio <0.5, or atypia explained by benign cause (polyomavirus, stones, conduit)
Atypical urothelial cells	N:C ≥0.5 + ≥1 feature (hyperchromasia, coarse chromatin, irregular nuclear membrane)
Suspicious for HGUC	N:C ≥**0.7** + ≥2 features, few cells (<5–10)
HGUC	Same, many cells

ALGORITHM
Positive high-grade cytology + negative cystoscopy → random bladder and prostatic urethral biopsies → selective upper-tract washings for cytology ± ureteroscopy

Urine biomarkers

More sensitive but less specific than cytology; **none replaces cystoscopy**. A negative marker never excludes cancer (≥10% residual risk in gross haematuria).

Assay	Target	Note
Nuclear matrix protein 22 (NMP22)	Nuclear structural protein (lab or point-of-care)	Qualitative: high specificity
Bladder tumour antigen (BTA) stat/TRAK	Complement factor H (related protein)	False positives with haematuria (serum cross-reactivity)
UroVysion fluorescence in situ hybridisation (FISH)	Aneuploidy chromosomes 3, 7, 17; 9p21 loss	Reflex after atypical cytology; weak for low grade
ImmunoCyt	Carcinoembryonic antigen (CEA) + two mucins (cell based)	**Highest sensitivity**; operator dependent
CxBladder	Five-mRNA panel	Needs validation

6 · STAGING AND RISK GROUPS

Stage	Tumour, nodes, metastases (TNM), American Joint Committee on Cancer (AJCC) 2017
0a / 0is	Ta (noninvasive papillary) / Tis (CIS)

Stage	Tumour, nodes, metastases (TNM), American Joint Committee on Cancer (AJCC) 2017
I	T1 – lamina propria (subepithelial connective tissue)
II	T2a inner / T2b outer half muscularis propria, N0
IIIA	T3a (microscopic perivesical) / T3b (macroscopic) / T4a (prostatic stroma, uterus, vagina) N0; or T1–T4a N1 (single true pelvic node)
IIIB	T1–T4a N2 (≥2 true pelvic nodes) or **N3 (common iliac)**
IVA / IVB	T4b (pelvic/abdominal wall) or M1a (distant nodes) / M1b (organs)

- **2017 change**: regional nodes without distant metastasis = stage III (previously IV). MCQ Common iliac = N3, not M1a.
- **T1a/T1b** (relative to muscularis mucosae) is recommended: ~50% of T1 are upstaged to ≥T2 at cystectomy; deeper invasion = worse survival.
- **Prostatic urethra**: mucosal CIS is not adverse but escapes intravesical therapy; only **prostatic stromal invasion is T4a** (5-year survival historically <25%).

NMIBC risk group	Definition
Low	Primary, solitary, low-grade Ta, <3 cm
Intermediate	Low-grade Ta, multiple and/or recurrent and/or >3 cm
High	Any high grade, CIS or T1

Management: NMIBC → *Ch 130*; MIBC → *Ch 131*; TURBT technique → *Ch 132*.

7 · MCQ ONE-LINERS

1. Commonest presentation: painless gross haematuria (85%).
2. Smoking accounts for 30–40% of UC; risk near never-smokers after >15 years.
3. Slow NAT2 acetylators who smoke have highest risk.
4. Cyclophosphamide carcinogen: phosphoramide mustard; risk above 20 g cumulative.
5. Only S. haematobium is linked to bladder SCC.
6. Radiation-associated bladder cancer latency: 15–30 years; IMRT carries little risk.
7. Leukoplakia of bladder is not premalignant.
8. Cystitis glandularis may progress to adenocarcinoma.
9. Dysplasia with prior bladder cancer progresses to CIS in 60%.
0. CIS lacks FGFR3 mutations; carries TP53, RB, PTEN alterations.
1. RAS mutations are mutually exclusive with FGFR3.
2. Basal/squamous is the commonest consensus class (35%); neuroendocrine-like has worst survival.
3. Luminal markers: GATA3, KRT20, PPARG, FOXA1; basal: CK5/6, CK14.
14. Any micropapillary component → manage as micropapillary; early cystectomy even for NMIBC.
15. Plasmacytoid variant: CDH1 loss, intraperitoneal spread, chemosensitive.
16. Small cell bladder cancer: chromogranin/synaptophysin positive; cisplatin–etoposide first.
17. Preoperative radiation worsened survival in bladder SCC.
18. Bladder adenocarcinoma: exclude secondary source – colonoscopy.
19. Urachal carcinoma is staged by Sheldon; treated by en bloc dome, urachus and umbilicus.
20. Paris System HGUC: N:C ratio ≥0.7 with ≥2 features.
21. UroVysion FISH: chromosomes 3, 7, 17 and 9p21.
22. BTA false positive with haematuria (complement factor H in blood).
23. Highest-sensitivity urine marker: ImmunoCyt.
24. AJCC 2017: N1–N3 without M1 is stage III.
25. Only prostatic stromal invasion makes T4a.

130 Management Strategies for Non–Muscle-Invasive Bladder Cancer

Reading time ≈ 38 min (slow pace)

1 · DEFINITION AND NATURAL HISTORY

- **Non–muscle-invasive bladder cancer (NMIBC)**: urothelial carcinoma (UC) not invading detrusor: Ta, T1 and carcinoma in situ (CIS). Forms **70–80%** of new bladder cancers: Ta 60–70%, T1 20–30%, CIS ~10%. Pathology, grading and staging → *Ch 129*.
- Recurrence **40–60%**; progression from ~6% (low-grade [LG] Ta) to 17% (high-grade [HG] T1).
- **Grade, not stage, is the strongest predictor of progression**; CIS is second. HG Ta and HG T1 progress at similar rates. Primary CIS progresses in **>50%**; untreated CIS reaches muscle in 40–83%.
- **Understaging**: cystectomy for presumed NMIBC shows worse disease in roughly **one-third to one-half**; clinical T1 is unreliable.

> **KEY POINT** NMIBC is two diseases under one label. LG disease is a **recurrence** problem, and HG disease is a **progression** problem. Every decision follows from which one the patient has: chemotherapy treats recurrence, bacille Calmette-Guérin (BCG) immunotherapy treats progression, and cystectomy is for progression that cannot be controlled.

2 · RISK STRATIFICATION

Risk group drives the choice of intravesical therapy, the surveillance schedule and counselling. It is **dynamic** and must be reassessed at every recurrence.

American Urological Association/Society of Urologic Oncology (AUA/SUO) 2016 (updated)

Risk	Criteria	Intervention
Low	LG solitary Ta **≤3 cm**; papillary urothelial neoplasm of low malignant potential (PUNLMP)	Single post-op chemotherapy instillation
Intermediate	LG Ta recurring within 1 yr; LG Ta solitary >3 cm or multifocal; **LG T1**; **HG Ta ≤3 cm**	6-week induction chemotherapy or BCG; maintenance to 1 yr
High	HG T1; recurrent HG Ta; HG Ta >3 cm or multifocal; **any CIS**; any BCG failure in HG; **any variant histology**; **any lymphovascular invasion (LVI)**; any HG prostatic urethral involvement	Induction BCG + maintenance to **3 yr**. BCG-unresponsive: radical cystectomy (RC), pembrolizumab, nadofaragene, gemcitabine–docetaxel

European Organisation for Research and Treatment of Cancer (EORTC) and Club Urológico Español de Tratamiento Oncológico (CUETO) scoring models

Factor	EORTC	CUETO
Number of tumours, prior recurrence rate, T stage, concurrent CIS, grade (World Health Organization [WHO] 1973)	✓	✓
Tumour diameter	✓	—
Age, sex	—	✓

- EORTC gives the probability of recurrence and progression at 1 and 5 years. **Recurrence** depends on prior recurrence rate, number and size; **progression** depends on T stage, CIS and grade.
- EORTC **overestimates** risk because its cohorts had little maintenance BCG, perioperative mitomycin C (MMC) or repeat transurethral resection of bladder tumour (TURBT). CUETO was derived in BCG-treated patients, so it fits current practice better. MCQ Both use **WHO 1973** grading.

3 · ENDOSCOPIC MANAGEMENT

TURBT: principles

- TURBT is both diagnostic and therapeutic (technique → *Ch 132*). Its two quality levers are **complete resection** and **detrusor in the specimen**. Missing muscle is linked to higher mortality.
- Assess the upper tracts at diagnosis with computed tomography (CT) urography, or retrograde pyelography at TURBT. Upper tract tumours occur in **<5%**; risk factors are trigonal tumour, CIS and high-risk disease.
- **Diverticular tumour**: there is no detrusor, so invasion beyond the lamina propria = perivesical fat = **T3a**. LG: cold-cup resection + fulguration. HG: strongly consider partial cystectomy or RC.
- **En bloc resection** (loop or laser) gives fewer perforations, more detrusor and better staging, but **no difference in recurrence**.

Repeat (second-look) TURBT

- Rationale: residual tumour at the resection site in **≥40%**. In HG T1, re-resection upstages to muscle invasion in **25–30%**, and in up to ~50% if the first specimen had no muscle. It changes treatment in one-third and lowers recurrence and progression in T1 (randomised data).
- **Residual T1 at re-resection → progression up to 80%** → consider timely cystectomy.

Recommended (within 6 weeks, include muscularis propria)	Consider
Incomplete initial resection; all T1 (especially without muscle); no muscle in specimen	High-risk HG Ta (multifocal and/or large)

Biopsies and office management

- Cold-cup biopsies avoid cautery artefact. Random biopsies are **not** indicated in low risk; consider them with positive cytology or high-risk history. Sample the prostatic urethra if there is CIS or a bladder neck/trigone tumour.
- Small (**≤1 cm**) LG recurrences: office fulguration or laser ablation. No tissue is obtained, so a **prior LG diagnosis and negative cytology are mandatory**.
- **Active surveillance** of small LG tumours is safe in selected patients.

Enhanced cystoscopy

	Blue-light (hexaminolevulinate, HAL)	Narrow-band imaging (NBI)
Principle	Porphyrins accumulate in tumour and fluoresce red under blue light	415 nm (superficial) and 540 nm (deeper) light absorbed by haemoglobin highlight vessels
Intravesical agent	Yes (catheter instillation)	No
Gain	~**one-third more CIS**; also smaller papillary tumours	More papillary tumours and CIS
Weakness	Cost; pragmatic randomised trial showed no 3-yr recurrence benefit	False positives ~60%

- HAL replaced 5-aminolevulinic acid because it avoids systemic photosensitisation. Progression benefit unproven. AUA: use when available.

4 · SINGLE POST-OPERATIVE INSTILLATION

- Purpose: kill floating tumour cells before they implant. It is an **anti-implantation** measure, not treatment of residual tumour.
- MMC given **within 6 h** reduces recurrence; given at 24 h it does not. Level 1 evidence, with the greatest benefit in an **initial solitary LG papillary tumour**. Incremental benefit in recurrent or multiple tumours is limited.

- Agents: MMC, epirubicin, pirarubicin, and **gemcitabine**, now the US standard (2 g in 100 mL for 1 h; Southwest Oncology Group [SWOG] S0337). Thiotepa is ineffective.
- **Withhold** with extensive resection or suspected perforation, because of chemical cystitis, skin desquamation, contracture and rare deaths.
- **Never give BCG immediately after transurethral resection (TUR)**: intravasation of live bacteria can cause sepsis and death.

5 · INTRAVESICAL CHEMOTHERAPY

- Reduces **recurrence** but has **no proven effect on progression**.
- **Intermediate risk**: single-agent MMC or gemcitabine is as effective as BCG, especially for LG multifocal recurrent disease. Give a 6-week induction, then monthly maintenance for 1 yr.

Agent	Class	Dose	Signature toxicity
Mitomycin C	Alkylating	40 mg/20 mL weekly ×6	Rash, severe skin reaction, contracted bladder 5%
Gemcitabine	Nucleoside analogue	2 g/100 mL	Mild; minimal absorption
Doxorubicin / epirubicin	Anthracycline	50 mg/50 mL	Chemical cystitis, contracted bladder
Valrubicin	Anthracycline	800 mg/75 mL	Approved by the Food and Drug Administration (FDA) for BCG-refractory CIS; complete response (CR) ~1/5
Thiotepa	Alkylating	60 mg/30 mL	Myelosuppression
Docetaxel	Taxane	75 mg/100 mL	Myelosuppression

- **Optimised MMC delivery** (Au 2001): drain residual urine, restrict fluids overnight, give oral sodium bicarbonate (reduces drug degradation), and concentrate to 40 mg in 20 mL.
- **Device-assisted MMC**: electromotive drug administration (EMDA) and chemohyperthermia at 42°C (Synergo; hyperthermic intravesical chemotherapy, HIVEC). These lower recurrence compared with passive MMC.
- **Gemcitabine–docetaxel (Gem/Doce)**: gemcitabine 1 g/50 mL for 60–90 min → drain → docetaxel 37.5 mg/50 mL for 60–120 min. Give weekly ×6, then monthly for 2 yr in responders. It is used for BCG-unresponsive disease, during shortages and increasingly in BCG-naive HG disease; a phase III trial against BCG is under way.

6 · BACILLE CALMETTE-GUÉRIN (BCG)

Mechanism

BCG (attenuated mycobacterium, the tuberculosis vaccine) binds **fibronectin** in the bladder wall, which triggers a local immune response. Cytokines are released and granulocytes, mononuclear cells and dendritic cells flood in, causing cell-mediated cytotoxicity. An effective response is **T-helper type 1 (Th1)**, with interferon-γ, interleukin (IL)-2 and IL-12; IL-10 drives a suppressive Th2 response. Release of tumour necrosis factor–related apoptosis-inducing ligand (TRAIL) correlates with response. No strain is superior. Advanced age and immunosuppression blunt the response.

Administration

ALGORITHM

Wait **2–4 weeks** after TURBT (re-epithelialisation) → urinalysis before each dose (exclude infection or gross bleeding) → 1 vial in 50 mL saline by catheter → retain 1–2 h, restricting fluids, diuretics and caffeine beforehand. If catheterisation is traumatic, delay several days.

Contraindications

Absolute	Relative
Immediately after TUR; gross haematuria; traumatic catheterisation; previous BCG sepsis; immunosuppression*; total incontinence	Urinary tract infection (UTI); liver disease (isoniazid could not be given for sepsis); previous tuberculosis; anti–tumour necrosis factor (TNF) drugs; poor performance status; advanced age

- Most absolute contraindications are a **breached urothelium → intravasation**. *BCG is in fact safe in transplant recipients and patients on steroids, but the response is weaker.

Efficacy

- BCG is the **only intravesical agent shown to reduce progression**, and the benefit is confined to trials using **maintenance**. BCG beats MMC only when maintenance is given.
- CIS initial CR up to **84%**; ~half are durable. Only ~30% are free of recurrence or progression at 10 yr, so close follow-up is mandatory.
- Persistent CIS after induction: a second induction gives a further 30–50% CR. Beyond 2 courses, progression risk is 20–50% (~7% more per extra course). **Response at 6 months predicts prognosis.**

Schedule

ALGORITHM
SWOG (Lamm 2000): induction weekly ×6 → 3 weekly instillations at **3 and 6 months, then every 6 months to 3 yr**. High risk: full dose for 3 yr. Intermediate risk: 1 yr (EORTC 30962).

- Only 16% completed the full SWOG schedule, yet maintenance was still superior. NIMBUS showed that **reducing instillation frequency fails**.

Toxicity management

Grade	Features	Management
1	Irritative symptoms, mild haematuria, fever <38.5°C, lasting <48 h	Anticholinergics, analgesics, NSAIDs (non-steroidal anti-inflammatory drugs); consider culture
2	Severe symptoms or >48 h	Culture, chest X-ray, liver tests; isoniazid + rifampicin + pyridoxine until resolution; resume at ½–⅓ dose
3	Haemodynamic change, persistent high fever; allergic (arthralgia, rash); solid-organ (liver, lung, kidney)	**Stop BCG**; isoniazid + rifampicin; add ethambutol; prednisone only **with** antimycobacterial cover; infectious disease input

- MCQ BCG is resistant to **pyrazinamide**. Avoid cycloserine (psychiatric effects). Asymptomatic granulomatous prostatitis needs no treatment.
- A quinolone for 24–48 h after each instillation reduces cystitis and treatment delays without loss of efficacy.

BCG shortage

ALGORITHM
Prioritise high risk → reduce dose, especially in maintenance → split vials → chemotherapy for intermediate risk (single agent) and high risk (Gem/Doce) → early RC for the highest risk.

- TICE is the only FDA-approved strain.

7 · RECURRENCE AFTER BCG

Category	Definition	Implication
Adequate BCG	≥5 of 6 induction + ≥2 of 3 maintenance, or ≥2 doses of a 2nd induction; for T1, ≥5/6 induction alone suffices	Threshold for "unresponsive"

Category	Definition	Implication
BCG-unresponsive	Persistent/recurrent CIS or HG Ta/T1 despite adequate BCG, or HG recurrence within **6 months (Ta/T1)** or **12 months (CIS)** of last dose	Unlikely to benefit from more BCG → strongly consider RC
BCG-resistant	CIS or HG Ta at the 3-month cystoscopy after induction	Reinduction/maintenance → CR 30–50%
BCG-intolerant	Inadequate BCG due to toxicity	Separate category; salvage chemotherapy
Delayed relapse	HG recurrence outside these windows	May respond to BCG rechallenge

> **KEY POINT** **Trap:** HG T1 at 3 months after induction alone already counts as unresponsive. CIS or HG Ta at 3 months does not; it is "resistant" and gets more BCG.

- After failure, check the prostatic urethra and upper tracts. BCG-unresponsive patients develop urethral or upper tract recurrence in ~half on long-term follow-up.

Bladder-sparing options (unfit for or declining RC)

Agent	Mechanism / schedule	CIS CR 3 mo → 12 mo	Cystectomy-free
Pembrolizumab (KEYNOTE-057)*	Anti-programmed death-1 (PD-1); 200 mg intravenously every 3 weeks up to 2 yr	41% → 19%	63%; grade 3–4 toxicity 13% (immune-related)
Nadofaragene firadenovec*	Replication-incompetent adenovirus carrying the interferon-α2b gene + Syn3 surfactant; intravesical every 3 months	53% → 24%	71%; grade 3–4 4%
N-803 + BCG	IL-15 superagonist → CD8+ T-cell and natural killer cell activation	71% (any time) → 45%	~85%
Gem/Doce	Weekly ×6, monthly ×2 yr	— → 60% (43% at 2 yr)	84%; HG recurrence-free survival (RFS) 65%/52% at 1/2 yr; no level 1 evidence

*FDA-approved. In every option, early CR roughly halves by 12 months, and none matches cystectomy for durability. Others: Gem/MMC, TAR-200, cretostimogene, valrubicin. Radiotherapy is largely palliative and has no effect on CIS.

8 · EARLY RADICAL CYSTECTOMY IN HIGH-RISK NMIBC

- Rationale: T1 failing BCG at 3 months progresses in **82%**; CIS failing BCG progresses in ~50%. With prolonged intravesical therapy, only ~¼ are alive with an intact bladder at 15 yr. Immediate RC for clinical T1 gives 10-yr disease-free survival of **92%**, falling to 64% if the disease was understaged. Even a 12-week delay worsens survival.
- **Patients who progress to muscle invasion under surveillance fare worse than those who present with muscle-invasive disease.**

ALGORITHM

Strongly consider RC for: BCG-unresponsive HG disease · persistent HG T1 on re-resection · T1 with CIS, LVI or variant histology · deep lamina propria invasion · diffuse CIS · diverticular HG tumour · distal ureteric or prostatic ductal involvement · tumour too large or inaccessible to resect completely · patient preference.

9 · SURVEILLANCE

Cystoscopy + cytology is the gold standard. Recurrence is best predicted by **tumour at the 3-month cystoscopy** and the number of tumours at first resection.

Risk	Cystoscopy	Cytology	Upper tract imaging
Low	3 months → annually from 12 months; consider stopping at **≥5 yr**	Consider	Only if haematuria
Intermediate	3–6-monthly for 1–2 yr → 6–12-monthly for 2 yr → annually	Consider	Consider every 1–2 yr
High	**3–4-monthly for 2 yr** → 6-monthly for 2 yr → annually **for life**	Same schedule	Annually for 2 yr, then lengthen

- Restart the clock at each recurrence. Base the schedule on the risk of the **most recently resected** tumour.
- **Cytology** has high specificity but low sensitivity. Persistent positive cytology with negative cystoscopy and imaging needs workup: biopsies, prostatic urethra and upper tracts.
- **Markers**: most are more sensitive but less specific than cytology. **UroVysion fluorescence in situ hybridisation (FISH)** (chromosomes 3, 7, 17, 9p21) is the most specific; an "anticipatory positive" can precede visible tumour. The AUA advises against using markers to replace cytology; use them to judge response to BCG and to resolve atypical cytology.
- **Extravesical**: upper tract UC is <1% in low risk and ~10% in high risk (13–18% after BCG), and the risk persists ≥15 yr. Prostatic urethral involvement in high risk is 20–40% at 10 yr; HG ductal involvement → cystoprostatectomy ± urethrectomy.

10 · PREVENTION

- **Smoking cessation counselling** is key, although its effect on recurrence is uncertain. Also advise increased fluid intake and a low-fat diet.

11 · LANDMARK TRIALS

Trial	Design	Result	Take-home
SWOG S0337	Immediate gemcitabine 2 g vs saline after TURBT for suspected LG	4-yr recurrence 35% vs 47% (LG 34% vs 54%); no extra toxicity	Gemcitabine is the standard single instillation
SWOG maintenance (Lamm 2000)	Induction ± maintenance BCG, recurrent Ta/T1/CIS	Median RFS 76.8 vs 35.7 months; only 16% completed	Maintenance is essential
EORTC 30962	Full vs ⅓ dose; 1 vs 3 yr maintenance	Toxicity stops ~7–8%	Intermediate: full dose 1 yr; high: full dose 3 yr
NIMBUS	Reduced-frequency (9 doses) vs standard BCG, HG	Recurrence doubled; stopped early	Do not reduce frequency
KEYNOTE-057	Single-arm pembrolizumab, BCG-unresponsive CIS	CR 41% (3 mo), 19% (12 mo)	First systemic therapy approved for NMIBC

12 · MCQ ONE-LINERS

1. Strongest predictor of NMIBC progression: grade; second: CIS.
2. LG T1 and HG Ta ≤3 cm are AUA intermediate risk.
3. Any CIS, LVI or variant histology is high risk.
4. EORTC and CUETO tables use WHO 1973 grading.
5. Re-TURBT within 6 weeks for all T1 and for incomplete resection.
6. HG T1 without muscle in the specimen: ~50% upstaged at re-resection.
7. Residual T1 at re-TURBT: progression up to 80%.
8. Diverticular tumour beyond the lamina propria = T3a.
9. Blue-light cystoscopy: greatest gain is CIS detection (~one-third more).
10. NBI wavelengths: 415 and 540 nm; false positives ~60%.
11. Single instillation: within 6 h; ineffective at 24 h.
12. US standard single instillation: gemcitabine 2 g in 100 mL (SWOG S0337).
13. Never give BCG immediately after TURBT: risk of septic death.

4. BCG binds fibronectin; an effective response is Th1.
5. BCG is the only intravesical agent that reduces progression.
6. SWOG maintenance: 3 weekly doses at 3 and 6 months, then 6-monthly to 3 years.
7. BCG is resistant to pyrazinamide.
8. Prednisone for BCG sepsis only with antimycobacterial cover.
9. Valrubicin: FDA-approved for BCG-refractory CIS; CR ~20%.
10. Adequate BCG: ≥5/6 induction + ≥2/3 maintenance (T1: induction alone).
11. BCG-unresponsive window: 6 months for Ta/T1, 12 months for CIS.
22. CIS at 3 months after induction = BCG-resistant → reinduce.
23. Pembrolizumab in BCG-unresponsive CIS: CR 41% at 3 months, 19% at 12 months.
24. Nadofaragene: adenovirus carrying interferon-α2b gene, every 3 months.
25. N-803 is an IL-15 superagonist given with BCG.
26. T1 failing BCG at 3 months: 82% progression.
27. Progression under surveillance does worse than de novo muscle-invasive disease.
28. High-risk cystoscopy: 3–4-monthly for 2 years, then lifelong.
29. UroVysion FISH targets chromosomes 3, 7, 17 and 9p21.

131 Management of Muscle-Invasive and Metastatic Bladder Cancer

Reading time ≈ 44 min (slow pace)

1 · NATURAL HISTORY AND HISTOLOGY

- Of new bladder cancers, **20–30%** are muscle-invasive bladder cancer (MIBC) and **~12%** node-positive or metastatic.
- **Primary MIBC** = invasive at diagnosis; **secondary** (**~20%**) = progressed from non-muscle-invasive bladder cancer (NMIBC). Secondary MIBC does worse: more understaging and lower pathological response to neoadjuvant chemotherapy (NAC), so it is biologically different.
- Most deaths after adequate local therapy occur within **2 years** and are systemic. They come from micrometastases already present at diagnosis, which is why every curative pathway wraps local treatment in systemic therapy.
- Urothelial carcinoma (UC) = 90%. **Radical cystectomy (RC) is standard whatever the histology.** Subtype changes only the use of NAC and radiation. Variants are upstaged more often.

Variant	Key points	Treatment
Squamous cell (5%)	Schistosomiasis, chronic cystitis	Poor chemo- and radiosensitivity → up-front RC + lymphadenectomy
Adenocarcinoma (2%)	Commonest tumour in exstrophy; urachal; exclude colorectal or breast metastasis	RC (urachal: partial cystectomy)
Neuroendocrine (small cell)	High stage; ectopic adrenocorticotropic hormone (ACTH), hypercalcaemia	**Chemosensitive** → NAC with etoposide + cisplatin, then RC
Micropapillary	Frequent lymphovascular invasion	Predominant → up-front RC
Mixed UC + variant	Quantify the variant component	UC-predominant → NAC; variant-predominant → RC

2 · CLINICAL STAGING

Sequence: depth of invasion on transurethral resection (TUR) → bimanual examination under anaesthesia (EUA) → computed tomography (CT) of abdomen and pelvis with and without contrast, plus excretory phase → chest imaging. In cT2, pathology upstages **40–54%** and downstages ~17%. Upstaging is predicted by lymphovascular invasion, variant histology, female sex and secondary MIBC.

Transurethral resection of bladder tumour (TURBT)

- Resect all visible tumour when safe, with a separate deep specimen. This lowers burden, may aid NAC response and is needed for bladder preservation; pT0 at RC improved survival even without NAC.
- High-grade T1 or no muscle in the specimen → repeat TUR within **6 weeks**. Without muscle, >40% are upstaged.
- **Prostatic urethral biopsy** (risk: trigone or bladder-neck tumour, carcinoma in situ [CIS], multifocal disease). Take a full loop from mid-prostate to the verumontanum at **5 and 7 o'clock**, where prostatic ducts are densest and CIS is commonest. Stromal involvement affects the need for NAC, urethrectomy and the diversion choice. Full thickness shows mucosa, ducts and stroma for accurate staging.
- **Frozen section at cystoprostatectomy**: positive TUR biopsy → frozen urethral margin is mandatory, and a neobladder is possible only if it is negative. Negative TUR biopsy → still do it if a neobladder is planned (higher negative predictive value).

Bimanual EUA

Examine before and after resection with the bladder empty: one hand suprapubic, fingers in the rectum or vagina. It both overstages and, more often, understages.

cT	Bimanual finding
T2a	Nonpalpable
T2b	Induration, no three-dimensional mass
T3a	Mobile three-dimensional mass
T4a	Invades prostate, vagina, rectum
T4b	Fixed to pelvic sidewall

Imaging and bloods

- **Image before TUR**: post-resection inflammation mimics T3. CT nodal sensitivity is only **30–50%**. Hydronephrosis strongly suggests extravesical disease.
- Magnetic resonance imaging (MRI): for poor renal function or contrast allergy; best local T-staging.
- **Vesical Imaging-Reporting and Data System (VI-RADS)** grades the low-signal muscularis line on MRI: intact in 1–3, interrupted in 4, breached into fat in 5. A score of **≥3** is suspicious for MIBC, so "equivocal" counts as positive.
- ^{18}F-fluorodeoxyglucose positron emission tomography (FDG PET)/CT is not routine. The American Urological Association (AUA) uses it when conventional imaging is abnormal and biopsy is not feasible. It is useless for local staging because FDG is excreted in urine.
- Chest CT for all; bone or brain imaging only if symptomatic or alkaline phosphatase raised.
- Renal function decides NAC and orthotopic diversion eligibility.

3 · PATHOLOGICAL STAGING (AMERICAN JOINT COMMITTEE ON CANCER [AJCC] 8TH EDITION) AND PROGNOSTIC MARKERS

Category	Definition
T2a / T2b	Inner / outer half of muscularis propria
T3a / T3b	Perivesical fat microscopic / macroscopic
T4a / T4b	Prostatic stroma, seminal vesicles, uterus, vagina / pelvic or abdominal wall
N1 / N2 / N3	Single true-pelvis node (perivesical now included) / multiple / **common iliac**
M1a / M1b	Nodes beyond common iliacs / non-nodal
Stage II / IIIA / IIIB	T2N0 / T3–T4a N0 or T1–T4a N1 / N2–N3
Stage IVA / IVB	T4b any N, or M1a / M1b

- Traps: common iliac = **N3, not M1**; T4b N0 is already stage IVA. pT4a needs **stromal** invasion; prostatic urethral or ductal CIS does not upstage. Organ-confined = ≤pT2bN0.
- **Molecular subtypes**: *luminal* (activating fibroblast growth factor receptor 3 [FGFR3] mutations) has the best survival. *Basal* (squamous or sarcomatoid) does worst but **gains most from NAC**. *p53-like* is chemoresistant. Subtypes do not yet guide treatment.
- **Circulating tumour DNA (ctDNA)**: more than a third remain positive after RC and do badly. In IMvigor010, adjuvant atezolizumab helped only ctDNA-positive patients.

4 · RADICAL CYSTECTOMY: PRINCIPLES AND EXTENT

- **RC + bilateral pelvic lymph node dissection (PLND) is the standard of care for resectable MIBC**, ideally after cisplatin-based NAC.
- Delaying RC more than **12 weeks** after diagnosis leads to more extravesical and nodal disease and worse survival.

- **Male**: bladder, perivesical tissue, prostate, seminal vesicles. **Female** (anterior exenteration): bladder, urethra, uterus and cervix, ovaries, tubes, anterior vaginal wall. Operative steps → *Ch 132*; diversion → *Ch 133–135*.
- **Male organ-sparing** (nerve, capsule, prostate or seminal-vesicle sparing) needs no bladder-neck disease and negative prostatic urethral and prostate biopsies. Incidental prostate cancer is found in a quarter to half of specimens, so reserve it for selected men.
- **Female organ-sparing**: with no tumour at the base or trigone, the uterus and anterior vagina may be spared. Benefit: better neobladder support, less prolapse. Preserve ovaries before menopause unless there is a breast or ovarian cancer risk. Keep the urethra unless the bladder neck is involved.

> **KEY POINT** An organ is spared only where it is oncologically silent. In men the gatekeepers are the prostatic urethra and bladder neck; in women, the bladder neck, base and anterior vaginal wall.

5 · PELVIC LYMPH NODE DISSECTION: EXTENT AND EVIDENCE

- About **25%** are node-positive at RC, and nodal status is the strongest predictor of survival. Recurrence is **70–80%** if node-positive vs ~30% if node-negative.
- Node-positive rate rises with T stage (≤pT1 1–7% … pT4 up to 80%).
- The obturator and internal iliac nodes are the sentinel stations. Skip metastases are rare, but common iliac nodes are involved in 7–13%.

Border	Standard PLND	Extended PLND
Medial / lateral	Ureter / genitofemoral nerve	
Caudal	Inguinal ligament, node of Cloquet	
Cranial	Common iliac bifurcation or ureteric crossing	Aortic bifurcation (to inferior mesenteric artery [IMA], controversial)

- **Node count**: a higher yield detects more nodal disease. About 25 nodes gives ~75% detection. A therapeutic threshold is unproven; 10 nodes is the usual pelvic-recurrence cut-off.
- Extranodal extension independently predicts death.
- **Randomised trials**: the German phase III and **Southwest Oncology Group (SWOG) S1011** (mostly post-NAC) both showed **no survival benefit**. S1011 found more grade ≥3 toxicity with extended dissection. **Standard PLND is the standard of care.**

6 · INTRAOPERATIVE DECISION MAKING

- **Clinically node-positive** → cisplatin chemotherapy first. If there is a radiological response, offer **consolidative RC**: 14–25% reach pathological complete response and survive much better.
- **Unexpected nodes at RC**: frozen section; proceed only if completely resectable and within the true pelvis.
- **Abort RC** if nodes are unresectable, there is extensive periureteral disease, the bladder is fixed to the sidewall or the rectosigmoid is invaded. Most then die within 2 years.
- **Ureteric margin**: the distal ureter is involved in 6–8%. A positive margin predicts upper-tract recurrence (UTR, 4–10%, mostly at 2–4 years). No guideline mandates frozen section. Send one for gross tumour; for CIS (skip lesions) resect maximally without sacrificing diversion length. Most positive margins can be cleared by sequential resection, but "converted" margins still carry higher UTR risk. Other UTR risks: prior bacille Calmette-Guérin (BCG), CIS, preoperative hydronephrosis.
- **Male urethra**: recurrence is 1–8%, mostly within 2 years. The highest risk is **prostatic stromal invasion** (up to ~30%). Consider urethrectomy for stromal invasion or diffuse prostatic urethral CIS. A positive pre-op prostatic biopsy does **not** bar a neobladder: **the intraoperative frozen urethral margin decides** (specificity ~100%, sensitivity variable). Low-grade papillary urethral tumour: resect pre-RC, keep urethra.

- **Female urethra**: involved in 2–12% (risk: bladder neck, base or anterior vaginal wall tumour). Urethrectomy is mandatory only for T4 involving urethra or vagina. Send a proximal urethral frozen section in every candidate for a neobladder.

7 · OUTCOMES AFTER RC

Stage	5-year disease-specific survival
≤pT1 / pT2 / pT3 / pT4	~90% / 70–81% / 40–52% / 16–44%
N0 / N+	62–80% / 21–35%

- **~50%** of RC patients ultimately die of bladder cancer, usually from metastases within 2 years.
- **Pelvic recurrence risk**: low (≤pT2) 5–8%; intermediate (pT3–4, negative margins, ≥10 nodes) 15–21%; high (pT3–4 with positive margins or <10 nodes) **39–46%**.

8 · NEOADJUVANT THERAPY

- **Cisplatin-based combination NAC for all eligible cT2–T4a N0 M0** (all guidelines). The National Comprehensive Cancer Network (NCCN) prefers dose-dense methotrexate, vinblastine, doxorubicin, cisplatin (**ddMVAC**) for 3–6 cycles; gemcitabine-cisplatin (**GC**) for 4 cycles is the alternative.
- Rationale: better tolerated than after RC, treats micrometastases early, downstages bulky tumours. **pT0 at RC** (5-year recurrence-free survival ~90%) shows that it worked. The downside is that non-responders' surgery is delayed.
- Single-agent cisplatin and carboplatin regimens give **no survival benefit**. Cisplatin-ineligible patients go straight to local therapy.

Criterion	Cisplatin-ineligible (any one; ~50% of patients)	Platinum-ineligible
Renal	Glomerular filtration rate (GFR) **<60 mL/min**	Creatinine clearance <30 mL/min
Eastern Cooperative Oncology Group (ECOG) performance status	≥2	≥2 (text) / ≥3 (table)
Hearing loss / neuropathy	Grade ≥2 / grade ≥2	— / grade ≥3
Heart failure	New York Heart Association (NYHA) class III–IV	NYHA III–IV; multiple high-grade comorbidities

Neoadjuvant immunotherapy (IO)

- Tumour programmed death ligand-1 (PD-L1) binds programmed death-1 (PD-1) on T cells to switch off killing; blockade restores it. Anti-PD-L1: atezolizumab, durvalumab, avelumab. Anti-PD-1: pembrolizumab, nivolumab.
- Phase II: **ABACUS** (atezolizumab) pathological complete response 31%; **PURE-01** (pembrolizumab) pT0 42%.
- In the source, neoadjuvant IO is not yet recommended because the phase III trials (NIAGARA, KEYNOTE-866, ENERGIZE) were pending. †NIAGARA has since reported improved event-free and overall survival with perioperative durvalumab + GC.
- Neoadjuvant radiation: downstages, no survival gain → not used.

9 · ADJUVANT THERAPY

- Target group: **pT3–4 or node-positive** at RC. Selection is on true pathological stage, but post-RC morbidity prevents delivery in about a quarter. Neoadjuvant is preferred: better compliance and level 1 evidence, whereas adjuvant trials were small and closed early.
- **Chemotherapy**: the Advanced Bladder Cancer (ABC) individual-patient meta-analysis (2022) showed a 6% overall survival (OS) gain at 5 years, greatest for N+ and extravesical disease. It is

not standard of care, but discuss adjuvant ddMVAC or GC in **cisplatin-naive** pT3–4/N+ patients.

- **Nivolumab for 12 months (CheckMate 274)** is offered for ≥pT3 or N+ disease after NAC, or in patients cisplatin-ineligible or declining chemotherapy. IMvigor010 (atezolizumab) was negative (included pT2).
- **Radiation**: pT3–4 local failure is 20–40%, and median survival after pelvic failure is ~9 months. Cairo trials improved pelvic control (mostly squamous). May consider for pT3/N+.

10 · BLADDER PRESERVATION

Trimodality therapy (TMT)

- TMT = **visibly complete TURBT + radiosensitising chemotherapy + radiation**. Concurrent chemotherapy beats radiation alone in two randomised trials.
- **Ideal candidate**: cT2, small solitary tumour, **no hydronephrosis, no CIS**, no nodes, good bladder function, motivated patient. Only 6–19% of MIBC qualify.
- Unfit for RC: looser criteria; extensive CIS and poor bladder are relative contraindications.
- Prerequisites: multidisciplinary team, a patient able to comply with frequent cystoscopy and imaging, and counselling about possible salvage RC. Offer surgically fit patients a balanced discussion of TMT vs RC ± perioperative chemotherapy.
- **Sensitisers**: mitomycin C + 5-fluorouracil (5-FU; **BC2001**: 2-year locoregional disease-free survival 67% vs 48%); cisplatin-based (Massachusetts General Hospital [MGH] and Radiation Therapy Oncology Group [RTOG] protocols); gemcitabine for cisplatin-ineligible patients; nicotinamide + carbogen (BCON).
- **Dose**: continuous course to **60–66 Gy** with cystoscopic restaging at 3–4 months (commonest). Split course (≈40 Gy → restage → consolidate) is historical (MGH/RTOG). **55 Gy in 20 fractions** is non-inferior.
- **Outcomes**: 80% of 5-year survivors keep their bladder. Propensity-matched RC vs TMT gave equivalent metastasis-free (~75%) and cancer-specific survival. **Salvage cystectomy is needed in 10–30%**.

> **KEY POINT** TMT contains cystectomy rather than replacing it. Salvage RC is the built-in safety net, so selection and lifelong cystoscopic surveillance decide success. Complete TUR cuts the need for cystectomy (~11% vs 42%).

Single modalities (none is primary therapy)

Modality	Use	Why it fails
Chemotherapy alone	Complete responders declining RC	Early intravesical recurrence (~half by 5 months); in RETAIN, half recurred despite DNA-repair mutations
Radiation alone (64–66 Gy/32 fractions or 55 Gy/20 fractions)	Unfit for surgery and chemoradiation; palliation	Inferior to chemoradiation; local control 27% in T3
Radical TUR	Superficial muscle invasion only	Understaging, ~25% occult nodes
Partial cystectomy + PLND	<3 cm solitary, no CIS, broad margin away from orifices; **urachal adenocarcinoma (treatment of choice)**	Recurrence, including late (lifelong follow-up). Routine frozen sections; convert to RC if margin positive; consider perioperative cisplatin

11 · METASTATIC UROTHELIAL CARCINOMA

- Bajorin adverse factors: Karnofsky performance status <80%, visceral metastasis (median survival 33, 13, 9 months with 0, 1, 2).
- **Cisplatin-eligible, first line**: GC or ddMVAC. Most respond but most progress. **GC = MVAC in efficacy** (von der Maase) and is less toxic. Dose-dense beat standard MVAC (European

Organisation for Research and Treatment of Cancer [EORTC] 30924).

- **Cisplatin-ineligible, platinum-eligible**: gemcitabine-carboplatin (EORTC 30986), or enfortumab vedotin (EV) + pembrolizumab (NCCN preferred). †EV-302 later showed EV + pembrolizumab roughly doubled OS over platinum chemotherapy in all comers, making it the preferred first line.
- **Platinum-ineligible**: pembrolizumab (KEYNOTE-052), or atezolizumab if PD-L1-positive (IMvigor210). Trap: **first-line IO alone only for platinum-ineligible patients.**
- **Maintenance**: **avelumab** if there is no progression after platinum (JAVELIN Bladder 100). IO + chemotherapy combinations (IMvigor130, KEYNOTE-361, DANUBE) are not recommended.
- **Second line**: **pembrolizumab** (KEYNOTE-045: OS 10.3 vs 7.4 months, regardless of PD-L1); alternatives nivolumab or avelumab. Atezolizumab failed (IMvigor211). Rechallenge with platinum if the last dose was ≥12 months earlier. Vinflunine is used in Europe only.
- **Antibody–drug conjugates (ADCs)** deliver cytotoxic drug selectively to an antigen. **EV** (targets Nectin-4, payload monomethyl auristatin E; neuropathy, rash) beat chemotherapy after platinum + IO (EV-301, OS 12.9 vs 9.0 months). **Sacituzumab govitecan** (targets trophoblast cell-surface antigen 2 [Trop-2], payload SN-38; neutropenia) is used third line.
- **Erdafitinib** (oral FGFR inhibitor) is for FGFR3 or FGFR2 alterations after platinum. Response 40%. Causes **hyperphosphataemia** and stomatitis.
- Failed: anti-vascular endothelial growth factor (VEGF) bevacizumab, sunitinib; anti-human epidermal growth factor receptor 2 (HER2) trastuzumab.
- **IO toxicity**: rash, colitis, hepatitis, pneumonitis, endocrinopathies (often **irreversible**); rarely myocarditis. Counsel; treat promptly.

ALGORITHM
Platinum-eligible → GC/ddMVAC (or gemcitabine-carboplatin; or EV + pembrolizumab) → response or stable disease → avelumab maintenance → progression → pembrolizumab, or erdafitinib if FGFR-altered → EV → sacituzumab govitecan. Platinum-ineligible → pembrolizumab → EV.

12 · FOLLOW-UP

- **After RC**: contrast CT with excretory phase plus chest CT every 6 months for 2–3 years, then yearly to 5 years (European Association of Urology [EAU], AUA; NCCN every 3–6 months for 2 years). Renal function every visit; vitamin B12 yearly from year 2 (bowel diversion). Urine cytology every 6–12 months.
- Urethral wash (controversial): for CIS or prostatic urethral tumour. Bloody urethral discharge → wash or flexible cystoscopy.
- Upper-tract recurrence (4–10%) is the commonest **late** recurrence; pelvic and distant recurrences come mostly within 2 years. Pelvic recurrence → chemotherapy and radiation; prognosis poor.
- Diversion checks: hyperchloraemic metabolic acidosis, renal function, stomal stenosis, parastomal hernia, stones → *Ch 133–135*.
- **After bladder preservation**: cystoscopy every 3–4 months in years 1–2, every 6 months in years 3–4, then yearly to **10 years**. CT urogram and chest CT every 3–6 months for 2 years, then yearly to 5. Biopsy any recurrence.

13 · LANDMARK TRIALS

Trial	Design	Result	Take-home
SWOG 8710	MVAC ×3 → RC vs RC	Median OS 77 vs 46 months; pT0 38% vs 15%	pT0 = surrogate; OS difference not significant (P = .06)
BA06 30894 (EORTC/MRC)	CMV (cisplatin, methotrexate, vinblastine) ×3 → RC/radiotherapy	10-year OS 36% vs 30%	Largest positive NAC trial
ABC meta-analysis	11 randomised NAC trials	OS HR 0.86; +5% at 5 years	Combination platinum only

Trial	Design	Result	Take-home
VESPER	ddMVAC ×6 vs GC ×4	Organ-confined 77% vs 63%; 3-year progression-free 66% vs 56%	ddMVAC better, more toxic
SWOG S1011	Limited/standard vs extended PLND	No difference in disease-free survival or OS	Standard PLND suffices
CheckMate 274	Adjuvant nivolumab vs placebo, pT3–4/N+	Disease-free survival 20.8 vs 10.8 months	Adjuvant standard
BC2001	Radiation ± mitomycin C/5-FU	Locoregional disease-free survival 67% vs 48%	Sensitise radiation
von der Maase	GC vs MVAC, metastatic	OS 14.0 vs 15.2 months (not significant)	Equal efficacy, GC less toxic
EORTC 30986	Gemcitabine-carboplatin vs MCAVI (methotrexate, carboplatin, vinblastine), cisplatin-ineligible	Equal OS, less toxicity	Gemcitabine-carboplatin standard
JAVELIN Bladder 100	Avelumab maintenance vs none after platinum	OS 21.4 vs 14.3 months (HR 0.69)	Maintenance standard
KEYNOTE-045	Pembrolizumab vs chemotherapy, second line	OS 10.3 vs 7.4 months	Second-line standard
EV-301	EV vs chemotherapy after platinum + IO	OS 12.9 vs 9.0 months	ADC third line

14 · MCQ ONE-LINERS

1. Secondary MIBC: about 20% of MIBC; worse outcome and lower NAC response.
2. Small cell bladder cancer → NAC with etoposide + cisplatin before RC.
3. Prostatic urethral biopsy sites: 5 and 7 o'clock beside the verumontanum.
4. VI-RADS ≥3 is suspicious for muscle invasion.
5. Common iliac nodes = N3; nodes beyond = M1a.
6. Prostatic urethral CIS does not make pT4a; stromal invasion does.
7. Basal subtype gains most from NAC; p53-like is chemoresistant.
8. Standard PLND cranial limit: common iliac bifurcation or ureteric crossing.
9. About 25 nodes gives ~75% chance of detecting nodal disease.
0. SWOG S1011: extended PLND gives no survival gain and more toxicity.
1. Male urethral recurrence: highest risk with prostatic stromal invasion.
2. Intraoperative frozen urethral margin decides orthotopic diversion.
3. Cisplatin-ineligible: GFR <60, ECOG ≥2, grade ≥2 hearing loss or neuropathy, NYHA III–IV.
14. ABC NAC meta-analysis: 5% absolute 5-year survival gain.
15. Adjuvant nivolumab (CheckMate 274) for one year in pT3–4 or N+ disease.
16. BC2001 sensitiser: mitomycin C + 5-fluorouracil.
17. TMT salvage cystectomy needed in 10–30%.
18. CIS and hydronephrosis are contraindications to TMT.
19. Partial cystectomy is the treatment of choice for urachal adenocarcinoma.
20. Avelumab maintenance after platinum response: JAVELIN Bladder 100.
21. Second-line pembrolizumab beat chemotherapy (KEYNOTE-045); atezolizumab did not (IMvigor211).
22. Enfortumab vedotin targets Nectin-4; sacituzumab govitecan targets Trop-2.
23. Erdafitinib signature toxicity: hyperphosphataemia.
24. Positive prostatic urethral biopsy does not preclude a neobladder if the frozen margin is negative.
25. Upper tract is the commonest site of late recurrence after RC.
26. Low-grade papillary urethral tumour: resect pre-RC; urethrectomy not needed.
27. Platinum rechallenge if last platinum was at least 12 months earlier.

132 Surgical Management of Bladder Cancer: Transurethral, Open and Robotic

Reading time ≈ 28 min (slow pace)

1 · FRAMING

- Transurethral resection (TUR) gives diagnosis and local stage in everyone, and is curative for low-grade non-muscle-invasive bladder cancer (NMIBC).
- Muscle-invasive bladder cancer (MIBC) needs radical cystectomy (RC) with pelvic lymph node dissection (PLND). Offer cisplatin-based neoadjuvant chemotherapy (NAC) to every eligible patient *→ Ch 131*.

2 · TRANSURETHRAL RESECTION OF BLADDER TUMOUR (TURBT)

Pre-operative evaluation

- Anticoagulants: **continue aspirin**. Hold warfarin, heparins, direct oral anticoagulants (DOACs) and P2Y12 inhibitors for a time set by each drug's half-life, with bridging agreed beforehand.
- Bimanual examination; digital rectal examination (DRE) if prostate suspected.
- Urine culture: treat **active infection before resection**.
- Upper tracts: computed tomography (CT) urogram, magnetic resonance imaging (MRI), or ultrasound + retrograde pyelogram. Iodinated contrast if glomerular filtration rate (GFR) **≥60** mL/min; gadolinium if GFR **>30**.
- Office flexible cystoscopy maps tumour location for planning. Tumour over a ureteric orifice → have ureteroscope, stent and fluoroscopy ready.

Set-up and inspection

ALGORITHM
Lithotomy → antibiotic within 1 h (avoid fluoroquinolones) → 30° lens with visual obturator (urethra, orifices) → 70° lens (bladder neck, dome, anterior wall) → 120° if still incomplete → record map → blue-light / narrow-band imaging where available (more tumours found, less recurrence; useful for carcinoma in situ (CIS))

	Monopolar	Bipolar
Irrigant	Sterile water or 1.5% glycine	0.9% saline
Electrolyte disturbance	Higher risk	Lower risk
Obturator reflex	Present	Reduced

Resection principles

- Aim: remove all visible tumour, with **muscularis propria (detrusor) in the specimen**. Use a loop electrode and cutting current.
- Tumour extends beyond its visible edge, so take a **~2 cm** margin of normal-looking tissue where possible.
- Then empty bladder; do examination under anaesthesia (EUA). A **mass still palpable after resection = cT3b**.
- † **En bloc resection** removes the tumour in one piece rather than in fragments. This preserves orientation and detrusor for better T1 staging.

Value of maximal resection	NMIBC	MIBC going to RC
Effect	Complete eradication improves outcome (therapeutic)	No survival gain from debulking. Residual tumour is prognostic only.
Why	Clinical staging is unreliable: many cT2 tumours are extravesical at cystectomy.	

Difficult locations

Site	Hazard	Countermeasure
Lateral wall	**Obturator jerk**: current stimulates the obturator nerve → leg adducts → loop is thrown laterally → perforation	Low distension; bipolar energy; muscle relaxant under general anaesthesia
Ureteric orifice	Coagulation seals the orifice → stricture, hydronephrosis	**Pure cutting current**, quick strokes. Routine stent not needed (stricture uncommon). A short-term stent after large resection or heavy cautery is reasonable.
Dome	Hard to reach; intraperitoneal perforation; heat injury to bowel even without perforation	Low distension; suprapubic pressure; straight loop; minimal cautery

Bladder perforation

	Extraperitoneal	Intraperitoneal
Site	Away from dome / posterior wall	Dome, posterior wall
Frequency	Commoner	Rare; large ones are morbid
Clues	—	Bladder will not distend; defect seen; abdomen distends
Confirm	Cystogram at the same sitting	
Management	**Foley catheter drainage + observation**	**Laparotomy**, inspect bowel carefully, repair; catheter + abdominal drain

> **KEY POINT** Perforation can follow an adequate resection of an advanced tumour. It **rules out immediate intravesical chemotherapy**, because the drug would leak extravesically.

Post-operative intravesical agents

These reduce **recurrence** but have little or no effect on **progression**. Give only if there is no perforation.

Agent	Setting / timing	Bottom line
Single-dose mitomycin C / epirubicin	NMIBC; **within 24 h**	Meta-analysis: ~**13%** absolute fall in recurrence
Gemcitabine 2 g in 100 mL saline (Southwest Oncology Group, SWOG S0337)	Suspected low-grade; immediately after TUR	12% absolute fall at 4 years; no grade 4–5 toxicity
Bacille Calmette–Guérin (BCG)	High-grade; most active agent	**Never immediately.** Wait **2–4 weeks** and confirm no gross haematuria († risk of systemic BCG absorption).

3 · RADICAL CYSTECTOMY: PREPARATION

- Staging: CT chest/abdomen/pelvis (MRI if renal function is poor).
- **Stoma marking for every patient**, sitting and standing, whatever diversion is planned. Conversion to a conduit is rare but must be consented.
- **No routine mechanical bowel preparation** and no oral antibiotics. Omitting it speeds bowel recovery and lowers wound complications.
- **Alvimopan** (peripheral μ-opioid antagonist) 30–90 min pre-op, then twice daily → less ileus, shorter stay.
- IV antibiotic within 1 h of incision covering skin flora and bowel gram-negatives/anaerobes, e.g. **cefoxitin**.
- **Venous thromboembolism (VTE) prophylaxis**: pneumatic compression + pharmacological prophylaxis started **before induction**. Extend for **4 weeks** (28 vs 8 days → major VTE reduced, no

extra bleeding). Enoxaparin (low-molecular-weight heparin, LMWH) is standard. A DOAC (apixaban) is as effective with better compliance, so consider it for the extended course.

- Position: men supine, table flexed ~15° at the anterior superior iliac spine. Women in low lithotomy for vaginal access. Prep xiphoid to thighs with 10% povidone-iodine; avoid chlorhexidine on genitalia.

4 · OPEN RADICAL CYSTECTOMY: TECHNIQUE

ALGORITHM
Lower midline incision; fascia opened exactly in the midline → Retzius space; bladder freed from sidewalls → urachus divided; peritoneum incised lateral to the medial umbilical ligaments → vas / round ligament divided → caecum and small-bowel root mobilised; left Toldt line incised → **retrosigmoid window** (route for the left ureter) → self-retaining retractor (check the vena cava is not compressed) → ureters freed with adventitia; obliterated umbilical artery divided first for length; ureters ligated and divided → PLND → vesical pedicles stapled, clipped or sealed

- **Distal ureteric frozen section** is controversial. A positive margin predicts upper-tract recurrence, but re-resecting to a negative margin neither prevents that recurrence nor proves a survival benefit.

Male

ALGORITHM
Peritoneum over seminal vesicles incised (keep a margin if tumour is at the base) → rectum separated; Denonvilliers fascia incised → posterior pedicles controlled → endopelvic fascia opened → dorsal venous complex ligated → urethra divided

- After radiation, advanced disease or fibrosis: **sharp dissection under vision**, not blunt.
- **Rectal injury → primary repair** ± flap coverage ± faecal diversion.
- **Urethral margin**: frozen section if a neobladder is planned. Negative frozen section has a very high negative predictive value; prostatic urethral biopsy predicts the margin poorly. **Positive margin → no orthotopic neobladder** (warn the patient beforehand).
- **Nerve sparing**: few men are potent at 1 year; age predicts recovery. Local recurrence has been reported even in pT2. Avoid with advanced disease or pre-existing erectile dysfunction.
- **Prostate sparing**: improves potency and continence, but occult prostate cancer is common and distant metastasis frequent. Requires normal DRE and prostate-specific antigen (PSA), transurethral prostatic resection and negative frozen section.

Female

- Standard operation = **anterior exenteration**. It is essential for a positive EUA or cT4a. Women present later and female sex independently predicts cancer death.
- Counsel about sexual morbidity (dyspareunia, dryness, low libido). Ovaries make testosterone after menopause, so discuss **ovarian sparing** when disease is not locally advanced.

ALGORITHM
Ovarian vessels ligated → posterior cervical fornix incised → lateral/posterior pedicles controlled → urethra excised from above or the introitus → **clamshell vaginal closure** (keeps girth, loses length), watertight, 2-0 polyglactin → vaginal pack out day 2

- Lateral vaginal pedicles are sinusoidal, so haemostasis must be meticulous. A leaky closure causes persistent peritoneal fluid drainage per vaginam.
- **Vaginal + urethral sparing for neobladder**: requires ≤cT2, uninvolved bladder neck and negative urethral frozen section. Develop the bladder–vagina plane on a pack or end-to-end anastomosis (EEA) sizer. Do not divide pedicles until mid-lateral vaginal wall. Preserve the striated sphincter.

5 · PELVIC LYMPHADENECTOMY

Template	Boundaries	Evidence
Standard	Lateral: genitofemoral nerve; medial: internal iliac artery; distal: Cooper ligament; proximal: ureter crossing common iliac	Reference
Extended	+ whole common iliac and presacral packets	Better staging; possible survival gain
Superextended	Up to inferior mesenteric artery (preaortic)	No benefit over extended

- Protect the obturator nerve; seal lymphatics at both ends.
- Node yield reflects surgical quality: **≥10 nodes** → better survival. In NMIBC at RC, yield >20 → better cancer-specific survival.
- A randomised trial showed **extended not superior to limited** dissection for recurrence-free or cancer-specific survival. The staging value is certain: it finds node-positive patients outside the true pelvis who then become candidates for adjuvant therapy.

6 · PARTIAL CYSTECTOMY

- **Indication: solitary, small MIBC without CIS**. Also used for **urachal adenocarcinoma**: en bloc umbilicus + urachus + dome with frozen-section margins.
- Same oncology as RC: PLND, NAC if cisplatin-eligible, and consent for RC if a clear margin cannot be achieved.
- Muscle-invasive recurrence → salvage RC.

ALGORITHM
PLND → cystotomy away from tumour → full-thickness excision with perivesical fat, 1–2 cm margin on frozen section → reimplant ureter if needed → 2–3-layer closure, test watertight → warm water irrigation (against seeding) → drain → cystogram day 7, then remove catheter

7 · ROBOTIC RADICAL CYSTECTOMY (RARC)

KEY POINT Bladder cancer has no effective salvage. Urine spillage, a poor PLND or a positive soft-tissue margin leads to almost uniformly fatal recurrence, so robotic technique is judged on oncology first.

- Indications match open surgery. Steep Trendelenburg + pneumoperitoneum strain heart and lungs; prior surgery and obesity are relative contraindications.
- **Docking**: da Vinci Xi side-docks from the patient's right (supine; lithotomy if vaginal access is needed). Older S/Si/X systems dock between the legs.
- **Ports** sit higher than for prostatectomy, to allow extended PLND and intracorporeal diversion. A **15-mm left assistant port** allows stapling and bagging whole node packets.

ALGORITHM
Sigmoid released → posterior plane over presacral packet → cul-de-sac / prostate apex → Retzius laterally → anterior pedicle clipped → ureters freed (**never grasp the ureter**) → **extended PLND with ureters intact** → ureters clipped; margin sent; left ureter passed under sigmoid → posterior pedicles → dorsal venous complex → urethra (clip proximally only if neobladder)

- PLND before ureteric division maintains urine output as a resuscitation guide and avoids prolonged ureteric occlusion.
- Remove each node packet whole and bag it immediately. The obturator nerve is **not** the posterior limit: continue down to the hypogastric vein.
- Female: EEA sizer in the vagina gives countertraction.

8 · COMPLICATIONS AND POST-OPERATIVE CARE

- Most patients have a complication but few are major. Gastrointestinal complications are commonest, then infection.
- **Ileus**: alvimopan, neostigmine (on telemetry), early feeding, opioid-sparing analgesia (ketorolac, paracetamol). **No routine nasogastric tube**.
- Continue VTE prophylaxis; early mobilisation; incentive spirometry.
- Ureteric devascularisation → ischaemic stricture (preserve adventitia). Vaginal leak/fistula → prevented by watertight closure. Rectal injury → repair ± diversion.
- Diversion-specific complications → *Ch 133–135*.

9 · LANDMARK TRIALS

Trial	Design	Result	Take-home
Grossman 2003	Randomised: NAC + RC vs RC	Median survival 46 → 77 months	Offer cisplatin NAC
SWOG S0337	Randomised: single gemcitabine vs saline after TUR	12% fewer recurrences at 4 years	Immediate gemcitabine for low-grade
RAZOR (2018)	Phase 3 non-inferiority: RARC vs open; open diversion	2-year progression-free survival 72.3% vs 71.6%; less blood loss, shorter stay	Robotic is oncologically equivalent
iROC	RARC + intracorporeal diversion vs open	More days alive and out of hospital at 90 days; fewer VTE and wound complications	Recovery advantage

10 · MCQ ONE-LINERS

1. Bipolar TURBT uses 0.9% saline; monopolar uses water or 1.5% glycine.
2. Obturator jerk: lateral wall tumour; prevent with low distension, bipolar energy, muscle relaxant.
3. Ureteric orifice resection: pure cutting current; routine stent unnecessary.
4. Palpable mass after complete TURBT = cT3b.
5. Extraperitoneal perforation → catheter drainage; intraperitoneal → laparotomy and repair.
6. Single post-TUR chemotherapy within 24 h cuts recurrence ~13%; not progression.
7. BCG: never immediately; wait 2–4 weeks without gross haematuria.
8. Maximal TURBT before cystectomy does not improve survival.
9. Iodinated contrast needs GFR ≥60; gadolinium contraindicated if GFR <30.
0. Routine mechanical bowel preparation not advised before RC.
1. Alvimopan: peripheral μ-opioid antagonist reducing post-cystectomy ileus.
12. Extended VTE prophylaxis for 4 weeks after RC; DOAC acceptable.
13. Standard PLND lateral limit: genitofemoral nerve; distal: Cooper ligament.
14. ≥10 nodes removed is associated with better survival.
15. Positive urethral frozen section contraindicates orthotopic neobladder.
16. Prostate-sparing RC risk: occult prostate cancer.
17. Female RC standard: anterior exenteration; female sex is an independent risk for death.
18. Postmenopausal ovaries still produce testosterone; consider sparing.
19. Female neobladder needs ≤cT2 and uninvolved bladder neck.
20. Partial cystectomy: solitary MIBC without CIS; cystogram day 7.
21. Urachal adenocarcinoma: en bloc umbilicus, urachus and dome.
22. RAZOR: robotic non-inferior to open for 2-year progression-free survival.
23. Obturator nerve is not the posterior limit of obturator dissection.

133 Use of Intestinal Segments in Urinary Diversion

Reading time ≈ 44 min (slow pace)

1 · SURGICAL ANATOMY

Stomach

- Coeliac trunk: left gastric (lesser curve); hepatic → right gastric + gastroduodenal → **right gastroepiploic**; splenic → short gastrics + **left gastroepiploic**. The gastroepiploic arcade on the greater curve carries the pedicle to the pelvis.
- Pedicle = whole antrum or a fundal wedge. Left gastroepiploic is occasionally atretic → use the right. A fundal wedge must **never reach the pylorus or lesser curve**. Antrectomy → Billroth I. On a left gastroepiploic pedicle, ligate short gastrics proximal to it and leave omentum on the vessels for support.
- Thick seromuscular layer peels off mucosa easily → suits a submucosal ureteric tunnel.

Small bowel

- ~22 ft; proximal 2/5 jejunum, distal 3/5 ileum. Jejunum: larger diameter, single arcades with larger vessels, thinner mesentery; ileum: multiple arcades, thicker mesentery.
- Do not rely on more than **8 cm** of bowel surviving away from a straight vessel → clear ≤8 cm of mesentery at a segment end. Base the pedicle on a palpable artery.
- **Pelvic (irradiated) small bowel – avoid after radiotherapy**: last 2 inches of terminal ileum (fixed) and the 5 ft starting ~6 ft from the ligament of Treitz (longest mesentery).

Colon

- Superior mesenteric artery (SMA): ileocolic (terminal SMA; last 6 inches of ileum + ascending colon), right colic, middle colic. Inferior mesenteric artery (IMA): left colic, sigmoid branches → superior haemorrhoidal; internal iliac → middle haemorrhoidal. All link through the **marginal artery of Drummond**.
- Mobilisation: ascending/descending colon along the avascular line of Toldt; transverse colon by dividing gastrocolic omentum, hepatocolic and phrenocolic ligaments.
- Right colic arising from ileocolic → distal ascending colon reaches the pelvis easily; from middle colic → may need division.

Watershed	Site	Between
Griffiths point	Splenic flexure	Middle colic ↔ ascending left colic
Sudeck's point	Rectosigmoid	Last sigmoid ↔ superior haemorrhoidal

> **KEY POINT** Watershed zones are fed by the smallest terminal branches of two arteries, so they become ischaemic first in hypotension or embolism → site anastomoses to one side of them.

2 · SEGMENT SELECTION

Depends on renal function, prior surgery/radiation, mesenteric mobility and diversion type. Neobladder reach (tension-free urethral anastomosis): caecal mobilisation, mesenteric Z-plasty, bed flexion.

Segment	Electrolyte effect	Advantages	Disadvantages	Choose when
Stomach	Hypochloraemic hypokalaemic alkalosis	Excretes H^+/Cl^-; less mucus; less permeable	Haematuria–dysuria syndrome, ulcers, gastric-surgery sequelae	**Renal failure**, short gut, dense adhesions
Jejunum	Hyponatraemic hypochloraemic **hyperkalaemic** acidosis	Outside radiation field	Severe electrolyte syndrome	Nothing else; use most distal

Segment	Electrolyte effect	Advantages	Disadvantages	Choose when
Ileum	Hyperchloraemic acidosis	Mobile, constant supply	B12, bile-salt, fat loss; obstruction ~10%	Default conduit; ureteric replacement
Colon	Hyperchloraemic acidosis	Large; easy tunnelled antireflux; transverse spared by radiation	Diarrhoea if ileocaecal valve used	Irradiated pelvis; antireflux wanted

- **Stomach**: gastrin normal/minimally raised; avoid antrum if fundus available; consider long-term H2 blocker/proton-pump inhibitor (PPI). **Haematuria–dysuria syndrome** (specific, common): bladder spasm, pain, haematuria/dysuria without infection, skin excoriation; intermittent; H2 blockers. Bacteriuria no lower than ileum/colon.
- Gastric-surgery complications – early: retention, haemorrhage, pancreatitis, duodenal leak; late: dumping, small-stomach syndrome, afferent loop syndrome, anaemia.
- Colon: obstruction needing reoperation ~5% (half the ileal rate).

Segment lost	Consequence
Ileum	Vitamin B12 malabsorption (anaemia, neuropathy); bile-salt loss → diarrhoea; loss of ileal brake → fatty diarrhoea
Ileocaecal valve	Bacterial overgrowth of ileum; fat/bile-salt malabsorption; vitamin A, D deficiency (osteomalacia); fast transit, bicarbonate and fluid loss
Jejunum	Fat, calcium, folate malabsorption (rarely relevant)
Colon	Diarrhoea, bicarbonate loss, dehydration

- **B12**: liver stores last **3–5 yr**; low levels common, clinical deficiency rare. Sensitive markers: homocysteine, methylmalonic acid. Many give empirical parenteral B12 after 5–7 yr of ileal diversion.
- Clam cystoplasty → more bowel dysfunction (detrusor overactivity associates with irritable bowel) → warn.

3 · BOWEL PREPARATION AND ENHANCED RECOVERY

	Mechanical	Antibiotic
Reduces	Faecal amount → total bacteria	Concentration per gram (to ~10^2/g)
Spill	Less volume, same inoculum/mL	Smaller inoculum

- Bacterial load: stomach low; descending colon **10^{10-12}/g**.
- Colorectal meta-analyses: mechanical preparation does not reduce leak (two found more dehiscence). Caveats: intravenous antibiotics were given; urological surgery opens long segments into a sterile tract.
- Oral antibiotics: kanamycin (best single), neomycin + erythromycin, neomycin + metronidazole. Useless in obstruction. Adverse: diarrhoea, pseudomembranous colitis, candidiasis, malabsorption. Recent colectomy data: oral + mechanical → fewer surgical site infections (SSIs), leaks and *C. difficile*.
- **Intravenous antibiotic is the most important measure**: before incision (ideally 1–2 h), must cover **anaerobes** (e.g., third-generation cephalosporin).
- Authors: clear liquids, magnesium citrate and oral neomycin the day before. Wound protectors reduce surgical site infection.

Enhanced recovery after surgery (ERAS)

- Counselling, nutrition, standardised analgesia/anaesthesia, early mobilisation, **no routine bowel preparation**;

- Postoperative ileus (POI) – commonest complication of cystectomy; risk higher than colectomy (pelvic deperitonealisation, urine exposure, long surgery, older/smokers).
- ERAS does not raise complications/readmissions; stay benefit inconsistent, but opioid use, diet, intensive care unit (ICU) time and quality of life improve. Gum and metoclopramide help; early feeding does not prevent protein depletion.
- **Alvimopan**: peripheral μ-opioid antagonist (poor blood–brain barrier entry → analgesia preserved); 12 mg 30 min–5 h pre-op, then twice daily ≤7 days (15 doses); inpatient only.
- ERAS trials excluded radiation, immunosuppression, liver/renal dysfunction → for salvage or benign cystectomy (radiation cystitis, neurogenic bladder, fistula) authors use full mechanical + antibiotic preparation and slow diet.

4 · INTESTINAL ANASTOMOSES

Bowel complications are the commonest cause of early morbidity and mortality after diversion.

1. **Exposure**; wall off with Mikulicz pads; clear mesentery ~0.5 cm.
2. **Blood supply**: transilluminate; pink, bleeding edges; avoid tension, excess cautery, tight sutures.
3. **Prevent spillage**: strip contents, non-crushing clamps; irrigate segment until clear.
4. **Serosa-to-serosa, inverted**, watertight, tension-free.
5. **No strangulating sutures**.
6. **Mesentery aligned**, untwisted; defect closure debated.

- Breakdown: ischaemia, faecal sepsis, **drain on the anastomosis**, irradiated bowel (most lethal bowel complications occur in irradiated patients).
- Sutured = stapled when done well; stapled side-to-side ileoileal is most common.
- Leak presents late (2nd week). Fistulas (faecal/urinary) in early weeks → sepsis. Neobladder–vaginal fistula: continuous incontinence months later; vaginal-sparing cystectomy reduces it by separating suture lines.
- **Obstruction** ~10% (ileum/stomach), ~5% colon; half early; causes adhesions > recurrent cancer. Prevent: unirradiated well-vascularised bowel, close apertures, decompression, omentum over anastomosis. **Reperitonealise**: tack antimesenteric border to lateral peritoneum and mesentery to posterior peritoneum – an open space traps bowel. Pelvic floor: anterior exenteration → sigmoid + omentum; total exenteration (especially before radiotherapy) → polyglactin mesh covered with omentum.
- Stenosis: early = oedema (continue decompression) or technical error (reoperate); late = ischaemia or perienteric infection → chronic partial obstruction.
- **Ogilvie syndrome**: dilated caecum within 3 days; water-soluble enema excludes obstruction; rupture risk with caecum **>12 cm** or rising white count → neostigmine/colonoscopic decompression; surgery only for ischaemia, peritonitis or refractory cases.

5 · URETEROINTESTINAL ANASTOMOSES

- Principles: minimal ureteric mobilisation; preserve periadventitia (blood supply); spatulate; fine absorbable sutures, mucosa-to-mucosa; **bring bowel to ureter**; fix and retroperitonealise; soft Silastic stents (leak 3–5% → near zero; fewer strictures).

Technique	Type	Key feature	Stricture	Leak
Leadbetter–Clarke	Nonrefluxing	Colon; tunnel under taenia (2.5–3 cm), ureter "as a hammock"	8–14%	2.5%
Goodwin	Nonrefluxing	Transcolonic tunnel from inside (3–4 cm)	No reliable data	
Bricker	Refluxing end-to-side	Any segment; simple	4–22% (mean 6%)	~3%
Wallace	Refluxing end-to-end	Conjoined ureters (side-by-side, Y, head-to-tail)	**~3% (lowest)**	~2%

- **Leadbetter–Clarke steps**: oblique 2.5–3 cm taenial incision → raise mucosa off muscularis → excise distal mucosal button → spatulate 5–7 mm → mucosa-to-mucosa 5-0 polydioxanone → loosely close seromuscular layer over uncompressed ureter.
- Wallace: spatulate 1.5–2 cm; lowest complications, easy endoscopic access; tumour blocking both ureters exceedingly rare.
- Best antireflux: colon – Pagano; small bowel – Le Duc, serosal apposition.

Reflux versus nonreflux

> **KEY POINT** Conduit pressure is damped by ureteric peristalsis and not transmitted to the renal pelvis. Reflux harms the kidney only with impaired peristalsis plus infection or obstruction. Antireflux tunnels stricture more and do not stop pelvic colonisation, so a stented refluxing anastomosis is the default. Reflux also allows loopogram imaging of the upper tracts in urothelial cancer.

Complications

- Urinary fistula days 7–10 (3–9%) → periureteric fibrosis → later stricture.
- **Stricture**: ischaemia, leak, radiation, infection; lifelong risk → lifelong surveillance. Classic: **left ureter where it crosses the aorta under the IMA** (over-stripping, angulation).
- Open excision and reanastomosis most durable; endoincision ~50–60% at 2 yr; balloon not durable. Poor endourological outcome: stricture <1 yr post-op, **≥1.5 cm**, left-sided. Metal stents only for limited life expectancy.

6 · CONDUIT DIVERSION

- Indications: after cystectomy; diseased or dysfunctional bladder; unusable bladder before transplant.
- **Ileal conduit** – simplest, fewest early complications. Avoid in short bowel, inflammatory small-bowel disease, irradiated ileum.

ALGORITHM

Segment **10–15 cm**, 10–15 cm from ileocaecal valve → transilluminate, keep a major arcade, clear 2 cm of mesentery at ends (windows ~5 cm, not more) → ileoileostomy; segment caudal to it → close mesenteric window → flush until clear → right ureter found by mobilising caecum, left via line of Toldt and passed behind sigmoid mesentery → stented anastomoses (Yankauer sucker as passer) → fix base retroperitoneally in right lower quadrant → stoma

- Duplex ureter missed → intraperitoneal leak; implant separately or join and implant as one.
- Stomal bleeding ~10%; intractable → stomal/conduit varices of portal hypertension → transhepatic portosystemic shunt or embolisation.
- **Jejunal conduit**: 15–25 cm from Treitz, stoma left upper quadrant; only when ileum and colon unusable (extensive ileal irradiation; adhesions with absent colon); contraindicated in nutritional bowel disorders. Hyperkalaemic hyponatraemic acidosis → sodium chloride, bicarbonate, thiazide.

	Transverse	Sigmoid	Ileocaecal
Best for	**Extensive pelvic irradiation**; intestinal pyelostomy	Exenteration with colostomy (no bowel anastomosis)	Long ureteric replacement; valve reinforced against reflux
Supply	Middle colic	Sigmoid vessels	Ileocolic
Avoid	—	Sigmoid disease; ligated hypogastrics with rectum in situ; pelvic irradiation	—

- All colon conduits: avoid in inflammatory bowel disease, chronic diarrhoea. Transverse segment usually 15 cm, anchored to midline retroperitoneum.
- **Ileal vesicostomy**: ileum on cystotomy → stoma; neurogenic, best with detrusor–sphincter dyssynergia; reversible; urethral incontinence (women with overactivity).

- All conduits: stents out days 4–6; drain retroperitoneally **never intraperitoneally**.

Complication (%)	Ileal conduit	Colon conduit	Continent
Bowel obstruction	10	4	0.8
Ureterointestinal obstruction	8	Antireflux 20	4
Acidosis needing treatment	16	5	Ileum 8; colon 27
Renal deterioration	18	15	—

Gastric conduit: acidosis 0%.

7 · STOMAS

- Most stomal complications are technical; commonest problem after diversion.
- Siting: mark sitting and supine; **through the rectus belly** (lateral → parastomal hernia); ≥5 cm from incision; avoid creases, scars, umbilicus, belt line, irradiated skin; peak of fat roll.
- Aperture: excise a skin disc, **keep subcutaneous fat** (fills dead space), cruciate fascial incision directly beneath (no angulation), spread rectus → admits two fingers. Exit perpendicular; do not trim fat at the margin; fit appliance in theatre.
- **Nipple (rosebud)**: bring out 5–6 cm for a 2–3 cm spout; four-quadrant eversion sutures (bowel edge → seromuscular 3–4 cm back → subdermis, never full-thickness skin). Straighten a curved ileum by incising mesentery 1 cm from bowel. Seromuscular myotomies reduce retraction in obesity.
- **Flush**: full-thickness bowel to subdermis → 1-mm raised margin.
- **Loop end**: close distal end → loop drawn through rectus by tape, closed end cephalad → rod for 7 days → enterotomy four-fifths cephalad, caudal edge everted.

	Flush	Protruding
Use	Continent catheterisable	Appliance-wearing conduit
Advantage	Low profile	Less stenosis, better fit, fewer skin problems

	End stoma	Loop (Turnbull)
Stenosis	Higher	Lower
Parastomal hernia	1–4%	4–20%
Best for	Standard	**Obese** (thick, short mesentery)

- Complications: necrosis, bleeding, dermatitis, hernia (ileal conduit 2–6.6%), prolapse, retraction, stenosis (ileal 20–24%, colon 10–20%; paediatric catheterisable >50%). Skin lesions: irritative, erythematous-erosive, pseudoverrucous → acidic urine, fitted appliance.

8 · RENAL FUNCTION AND DETERIORATION

- Deterioration after conduits 10–74%; in normal units ileal conduit 18%. Long term most decline regardless of type (motility loss, infection, stones, less often stricture). About 6% of ileal conduit patients die of renal failure. Ureterosigmoidostomy: more sepsis and renal failure; leading cause of death after 15 yr.
- Need depends on bowel length and dwell time → continent > conduit. **Tubular acidification matters as much as glomerular filtration rate (GFR).**

Parameter	Test	Threshold for continent diversion
Screen	Creatinine + urine protein	Creatinine **<2.0 mg/dL**, no proteinuria → proceed
GFR	Inulin clearance	**>35 mL/min** (average tolerance >40)
Acidification	Ammonium chloride load	Urine pH **≤5.8**
Concentration	Water deprivation	Osmolality **≥600 mOsm/kg**

- Exception: retentive diversion in end-stage renal failure (children awaiting transplant).
- **Tests invalidated by bowel**: urine osmolality, urine pH, serum urea and creatinine (reabsorbed).

9 · METABOLIC COMPLICATIONS

Determinants of solute absorption: **segment, surface area, dwell time, urine solute concentration, renal function, pH**.

Segment	Syndrome	Na	K	Cl	pH	Associated	Treatment
Stomach	Hypochloraemic hypokalaemic alkalosis	—	↓	↓	↑	↑ Aldosterone; lethargy, respiratory insufficiency, seizures, arrhythmia	H2 blocker, PPI; arginine hydrochloride; remove segment
Jejunum	Hyperkalaemic hyponatraemic hypochloraemic acidosis	↓	↑	↓	↓	↑ Renin, angiotensin; lethargy, vomiting, dehydration	Intravenous saline, bicarbonate, thiazide; remove segment
Ileum/colon	Hyperchloraemic acidosis	—	↓	↑	↓	Total-body K depletion; hypocalcaemia; fatigue, anorexia	Alkali (citrates, bicarbonate); chlorpromazine, nicotinic acid

> **KEY POINT** Stomach secretes HCl (alkalosis, low Cl); jejunum secretes NaCl (low Na, Cl; high K); ileum/colon absorb NH_4Cl (high Cl). Potassium is low with stomach and ileum/colon – **only jejunum is hyperkalaemic**.

Stomach – metabolic alkalosis

- Usually trivial; severe in renal failure or dehydration but can occur with normal kidneys.
- Vicious circle: H^+ loss adds bicarbonate → vomiting → volume, Cl and K depletion → aldosterone ↑ → kidney cannot excrete bicarbonate.
- **Gastrin** correlates with bicarbonate; above **120 ng/L** small gastrin rises cause large bicarbonate rises. Distension stimulates gastrin → warn high-gastrin patients about dehydration and overdistension.
- Treatment: rehydrate + H2 blocker → arginine hydrochloride if life-threatening → omeprazole → remove segment.

Jejunum – jejunal syndrome

- Jejunum secretes NaCl and water, absorbs K^+ and H^+ → hypovolaemia → renin–aldosterone ↑ → kidney makes low-Na, high-K urine → steeper gradient in the conduit → cycle perpetuates.
- Worse with proximal segments and parenteral nutrition; untreated can be fatal.
- Treatment: sodium chloride rehydration, bicarbonate; long-term oral salt, thiazide.

Ileum and colon – hyperchloraemic metabolic acidosis

- Some degree in most; treatment needed in a minority. Worst with **ureterosigmoidostomy** (most patients) and colonic continent pouches (most Mainz pouch patients need alkali).
- **Mechanism**: NH_4^+ replaces Na^+ on the Na^+/H^+ antiport, coupled with Cl^-/HCO_3^- exchange → **ammonium chloride absorbed** in exchange for carbonic acid; NH_4^+ also enters via K^+ channels.
- Fatigue, anorexia, weight loss, polydipsia; can be fatal.
- Chronic urine exposure → villous atrophy, but transporters stay functional – risk persists.

Agent	Action	Note
Sodium bicarbonate	Alkali	Effective; gas
Sodium citrate + citric acid (Bicitra, Shohl's)	Alkali	Poor taste

Agent	Action	Note
Potassium + sodium citrate (Polycitra)	Alkali + K	When sodium load undesirable and K needed
Chlorpromazine 25–50 mg thrice daily	Inhibits cyclic AMP → blocks Cl transport	Limits, does not correct; tardive dyskinesia
Nicotinic acid 400 mg 3–4 times daily	Same	Avoid in peptic ulcer, liver disease; flushing

Potassium and magnesium

- Hypokalaemia: commoner in colonic diversions (ileum reabsorbs K, colon does not); renal K wasting. Severe → flaccid paralysis.
- **Trap**: correcting severe acidosis alone drives K into cells → profound hypokalaemia → replace K with bicarbonate.
- Also hypomagnesaemia, hypocalcaemia.

Altered sensorium

- Causes: **ammonia** (commonest), magnesium deficiency, drug intoxication.
- Ammoniagenic coma: mainly liver dysfunction + ureterosigmoidostomy (also ileal conduit). Treat: drain diversion (rectal tube/Foley) → oral neomycin → protein restriction → intravenous arginine glutamate 50 g → lactulose.

Drug toxicity

Drugs absorbed by gut and excreted unchanged by kidney are reabsorbed → toxicity: **phenytoin, some antibiotics, methotrexate**. Monitor levels; **drain continent pouches during chemotherapy**.

Osteomalacia

Mechanism	Explanation
Chronic acidosis	Bone buffers H^+, releasing calcium → hypercalciuria
Vitamin D resistance	Some fail to remineralise on alkali alone
Renal calcium loss	Reabsorbed sulphate inhibits tubular Ca reabsorption
Parathormone resistance	Altered calcium:parathormone ratio

- Commonest with ureterosigmoidostomy. Joint pain, proximal myopathy; alkaline phosphatase ↑, Ca and phosphate low/normal.
- Correcting base deficit >2.5 mEq/L prevents bone loss. Treat: alkali + calcium → active vitamin D → 1α-hydroxycholecalciferol.

Infection

- ~3/4 ileal conduit urines bacteriuric, often only distal. **Treat pure Proteus or Pseudomonas** (predict deterioration); observe asymptomatic mixed cultures.
- Bowel cannot inhibit bacterial proliferation like urothelium; distension promotes translocation.

Stones

- Mostly struvite/calcium phosphate. Risk: hyperchloraemic acidosis, pyelonephritis, urease organisms, hypercalciuria, **foreign bodies (staples, non-absorbable sutures)**, mucus.
- Colon conduit 3–4% < ileal conduit 10–12% < continent caecal reservoir ~20%.

Cancer

- **Ureterosigmoidostomy**: 6–29% (mean 11%) after **10–20 yr** latency; mostly adenocarcinoma at the ureterocolic junction where urothelium meets colon bathed in faeces.
- Defunctioned anastomoses still develop cancer → **excise, do not ligate and leave**.

Diversion (Kälble 2011)	Risk	Surveillance
Ureterosigmoidostomy	2.6%	Endoscopy from 5 yr
Cystoplasty	1.6%	—
Colonic neobladder	1.3%	
Ileocaecal pouch	0.14%	If symptomatic
Ileal neobladder / conduit	0.05% / 0.02%	None routine

10 · NEUROMECHANICAL ASPECTS

- Bowel configuration determines storage volume and contraction pattern.
- † Detubularisation and reconfiguration into a sphere disrupt coordinated peristaltic contractions and, by Laplace's law, increase volume and lower pressure – the basis of continent reservoirs and neobladders.

11 · LANDMARK TRIALS

Trial	Design	Result	Take-home
Alvimopan (Lee 2014)	Randomised double-blind vs placebo, radical cystectomy	Bowel recovery 5.5 vs 6.8 days; stay 7.4 vs 10.1 days; less POI	Use with opioid-based analgesia

12 · MCQ ONE-LINERS

1. Griffiths point: splenic flexure; Sudeck's point: rectosigmoid.
2. Do not clear more than 8 cm of mesentery from a small-bowel segment end.
3. Only jejunum causes hyperkalaemia.
4. Stomach: hypochloraemic hypokalaemic metabolic alkalosis.
5. Ileum/colon: hyperchloraemic hypokalaemic metabolic acidosis via ammonium chloride absorption.
6. Haematuria–dysuria syndrome: gastric segments; treat with H2 blockers.
7. Gastrin above 120 ng/L sharply raises bicarbonate.
8. Life-threatening alkalosis: arginine hydrochloride.
9. Jejunal syndrome: salt, bicarbonate, thiazide.
10. Chlorpromazine and nicotinic acid block chloride transport via cyclic AMP.
11. Correct acidosis alone → hypokalaemic paralysis; give potassium.
12. Commonest cause of altered sensorium: hyperammonaemia.
13. Methotrexate toxicity: drain continent pouch during chemotherapy.
14. Continent diversion: GFR >35, urine pH ≤5.8, osmolality ≥600.
15. Bowel invalidates urine pH, osmolality, urea and creatinine.
16. Osteomalacia prevented by correcting base deficit >2.5 mEq/L.
17. B12 liver stores last 3–5 years.
18. Wallace anastomosis: lowest stricture rate.
19. Left ureter stricture: under the IMA.
20. Most important prophylaxis: intravenous anaerobic-cover antibiotic before incision.
21. Alvimopan: peripheral μ-opioid antagonist, max 15 doses.
22. Loop (Turnbull) stoma: obese; more hernia, less stenosis.
23. Transverse conduit: irradiated pelvis.
24. Treat pure Proteus or Pseudomonas in conduit urine.
25. Ogilvie syndrome: caecum >12 cm → neostigmine.
26. Never place a drain on an intraperitoneal anastomosis.
27. Stoma aperture: keep subcutaneous fat; admit two fingers.
28. Ureterosigmoidostomy cancer: 10–20 years; excise old anastomosis.

134 Continent Urinary Diversion

Reading time ≈ 46 min (slow pace)

1 · DEFINITION AND BASIC PRINCIPLES

Continent diversion = urine storage plus volitional emptying, either per urethra (**orthotopic neobladder**) or by intermittent catheterisation of a stable channel (**continent cutaneous reservoir**). Orthotopic is now preferred (natural voiding, no appliance, fewer revisions); cutaneous is kept for pre-existing incontinence or urethral disease.

- **Principle 1 – reliable outlet:** orthotopic needs an adequate external sphincter and an unobstructed urethra; cutaneous needs a robust catheterisable efferent limb with a continence mechanism.
- **Principle 2 – low pressure:** complete **detubularisation** and folding into a sphere (Kock double-folding; S/W shapes). Cutting the circular muscle abolishes coordinated peristaltic pressure waves, and a sphere gives the greatest volume per surface area. † By Laplace's law (P = 2T/r), the larger radius of a sphere gives lower pressure for the same wall tension.
- **Principle 3 – capacity:** mature volume **≥500 mL**. 44 cm ileum double-folded holds <200 mL initially and stretches to 500–600 mL at low pressure within a year — a large initial volume is unnecessary. Ileum stretches more than colon.

> **KEY POINT** Continence is a pressure balance: low reservoir pressure against adequate outlet resistance. Detubularisation lowers the first; the sphincter or efferent valve supplies the second. Failure of either causes leakage. The original Camey I (tubular ileum) failed because its pressure was too high.

2 · PATIENT SELECTION

80–90% of men and ~75% of women undergoing radical cystectomy (RC) are medical candidates for a neobladder, yet uptake is low. AUA (American Urological Association) and EAU (European Association of Urology) guidelines: counsel every patient on all three options; refer if the surgeon lacks orthotopic experience.

Category	Orthotopic only	All continent diversions
Absolute	Invasive cancer at urethral margin; significant urethral stricture	Severe renal or hepatic dysfunction; physical/mental inability to self-catheterise
Relative	Poor external sphincter; palpable bladder-neck or anterior vaginal wall tumour; prostatic stromal invasion; irradiated sphincter	Frailty, comorbidity, poor cognition, social difficulty, prior radiation or bowel resection

ALGORITHM

Estimated glomerular filtration rate (eGFR) >40 → intact external sphincter → adequate bowel → not frail/no severe liver, cardiac or psychological disease → men: no cancer at distal prostatic urethra; women: no palpable anterior vaginal tumour or cervical invasion → willing to self-catheterise (both sexes) → offer orthotopic

- Warn that intraoperative findings force a change in up to **~10%**; agree the back-up diversion. Mark a conduit site in everyone; authors prefer the umbilicus for a continent cutaneous stoma.
- **Conduit site rules:** free of fat creases standing, sitting and lying; away from umbilicus and scars; as lateral as possible while still through the middle of the rectus — otherwise parastomal hernia risk rises.

Oncological factors — urethral recurrence

Feature	Men	Women
Dominant risk	Prostatic urethral involvement; worst with **stromal invasion** (usually node-positive → neoadjuvant chemotherapy)	**Bladder-neck tumour**; anterior vaginal wall invasion
Pre-op assessment	Transurethral resection (TUR) biopsy at 5 and 7 o'clock beside verumontanum — optional	Bimanual examination under anaesthesia; bladder-neck biopsy
Definitive test	Intraoperative **frozen section of urethral margin** (more predictive than TUR biopsy; AUA requires a negative margin)	
Carcinoma in situ (CIS) of bladder	Not predictive	
Skip lesions	—	None: negative bladder neck → negative urethra

- Risk rises with depth of prostatic involvement: none < CIS/mucosal/ductal < stromal invasion. The **neobladder itself is protective** (meta-analysis hazard ratio 0.44), possibly from continued urine flow.
- **TUR biopsy vs frozen section:** TUR misses much prostatic involvement and frozen section predicts recurrence better; authors do not repeat TUR solely to exclude prostatic disease.
- Omitting urethral frozen section is unvalidated; authors send a urethral margin from the final specimen.
- Prostatic involvement → urethral surveillance (wash cytology, urethroscopy, watch for discharge/spotting).
- Women: vaginal wall invasion always accompanied bladder-neck disease. Preserve the urethra if bladder-neck biopsy and proximal urethral frozen section are negative.
- Persistent prostatic urethral disease → cutaneous diversion + urethrectomy. Preserve the female urethra whatever the diversion (distal urethrectomy devascularises the clitoris).
- **Locally advanced or node-positive disease found at surgery is not a contraindication**: pelvic recurrence ~7% (13% if extravesical/N+) and rarely disables the neobladder. Exception: grossly positive margin at pubis, rectum or side wall. Many advanced patients are long-term survivors who benefit. Adjuvant radiotherapy (45–50.4 Gy) to a neobladder is well tolerated.

Patient-related factors

- **Renal function** — the key contraindication. Bowel absorbs urea, K^+, Cl^- and secretes Na^+, HCO_3^- → acid load → hyperchloraemic metabolic acidosis if kidneys are poor. Threshold: creatinine **<1.7–2.2 mg/dL** or eGFR **>35–40 mL/min**. Moderate impairment (eGFR 40–59) is safe: decline is no faster than with conduit. Obstructive uraemia often reverses after cystectomy. All patients: hydration, avoid nephrotoxins, oral sodium bicarbonate. Metabolic detail → *Ch 133*.
- **Age, obesity**: not contraindications. Elderly regain continence more slowly but ultimately similar. Commonest technical reason to abandon orthotopic = short/fatty mesentery that will not reach the urethra; obese patients may do better without a stoma through a thick wall. **Reach manoeuvres:** deeper mesenteric division at the distal resection site, discard a proximal bowel segment, reduce table flexion/Trendelenburg, gentle gradual traction to the urethra.
- **Self-catheterisation** needed in 10–20% of men and ~30% of women with neobladder and all cutaneous pouches; retention may appear years later, so every candidate must be willing. Failure to catheterise → perforation, stones, infection, renal decline. Multiple sclerosis, quadriplegia, frail/impaired → conduit (easier for carers).
- **Sphincter weakness**: man → artificial urinary sphincter; woman → Burch/sling (likely catheterisation). Severe stricture → no orthotopic.
- **Prior radiation**: neobladder feasible in selected patients without incontinence, stricture, fistula or severe bowel damage; prostate brachytherapy/high-dose radiation and cervical vaginal implants

damage the rhabdosphincter most. Cystoscope the sphincter pre-op, but the final decision is intraoperative.

- **Prior prostatectomy**: pre-cystectomy continence predicts outcome (poor → likely artificial sphincter). Flexible cystoscopy during apical dissection locates the old anastomosis so no bladder is left.
- **Prior bowel resection**: take down the old anastomosis and use it as one end of the reservoir (avoids an ischaemic segment); short bowel risk → sigmoid neobladder.

3 · CONTINENCE MECHANISM AND ITS PRESERVATION AT CYSTECTOMY

The striated **rhabdosphincter** (pudendal nerve) is the critical structure; it lies **anterior and lateral** to the urethra in both sexes (omega-shaped in men around the membranous urethra; mid-urethral continence zone in women).

Men — apical dissection

- Posterior dissection never distal to the apex. Nerve-sparing order: lateral pedicles → divide urethra → retrograde dissection off rectum and neurovascular bundles → posterior pedicles last.
- Incise endopelvic fascia minimally, only slightly incise puboprostatic ligaments, control dorsal venous complex (DVC) with **absorbable** suture (no clips/staples to erode into the anastomosis) and avoid deep bites into levators.

ALGORITHM
Divide anterior attachments → incise endopelvic fascia, sweep levators → ligate superficial dorsal vein → slightly incise puboprostatic ligaments → ligate DVC, divide close to apex, suspend to pubic periosteum → incise anterior urethra → place 2-0 sutures → divide catheter → posterior sutures through rectourethralis/Denonvilliers → divide posterior urethra → tag sutures until after ureteroenteric anastomosis

- Urethral sutures go in **before** complete division because the male urethra retracts.
- Apical frozen section: tumour → urethrectomy + cutaneous/conduit; atypia/focal CIS → second distal margin.
- Nerve sparing: continence benefit controversial; reasonable in potent men when safe. It preserves membranous urethral sensation, which triggers reflex sphincter closure (helps night continence).

Women

- Spare uterus, anterior vagina and ligaments in selected low-stage disease → no vaginal fistula, better sexual function, less late retention. Keep vaginal dissection no more dorsal than 2 and 10 o'clock (lateral neurovascular tissue).
- Uterus preserved: take bladder branches distal to the uterine artery. Dissect bladder off (not excise) the anterior vagina; a deep trigonal tumour may need partial vaginal wall excision (fistula risk ↑, not absolute contraindication). Palpable bladder-neck/vaginal tumour → en bloc urethrectomy + cutaneous diversion.

ALGORITHM
Posterior pedicles → sharp plane off anterior vagina → Foley balloon (~20 mL) marks junction → sweep fat off anterior urethra → free bladder neck and first 0.5 cm urethra → clamp bladder neck (prevents tumour spill) → divide just distal → frozen section

- Foley balloon marks the vesicourethral junction; leave endopelvic fascia untouched; divide just distal to the bladder neck (bladder neck not needed for female continence). The female urethra does not retract, so sutures go in after removal.
- Always interpose an **omental flap** between vagina and pouch (prevents fistula, supports pouch). Authors add sacrocolpopexy against prolapse and late retention. If anterior vaginal wall is excised, keep a rim proximal to the urethra so vaginal and urethral suture lines do not abut.

4 · CHOICE OF BOWEL AND URETERIC REFLUX

- Distensibility: ileal circular > colonic circular > ileal longitudinal > colonic longitudinal. **Ileum is the default**: best compliance, largest capacity, mobile mesentery; mucosa atrophies with urine exposure (less mucus and absorption). Colon: pressure waves, needs bigger initial volume — for diseased/resected ileum. Stomach: secretes H^+Cl^- (helps acidosis) but poor compliance, haematuria–dysuria; rarely used.
- Protect blood supply: avoid deep mesenteric incision at the proximal division (reservoir supply).
- Close with continuous absorbable suture — metal staples cause stones.
- **Antireflux is unnecessary** in a neobladder: Valsalva voiding raises pressure equally throughout, and every antireflux valve can obstruct. Direct (Wallace/Leadbetter) anastomosis has the lowest stricture rate (**3–6%**) vs Le Duc 7–29%. **Obstruction, not reflux, destroys kidneys.**
- Antireflux options when wanted: serous-lined extramural tunnel (needs long, non-dilated ureters) or T-pouch flap valve (accepts dilated/short ureters). Kock nipple failures are late (often >10 years) and silent until bilateral hydronephrosis.

5 · ORTHOTOPIC NEOBLADDER DESIGNS

Design	Bowel / shape	Ureters	Key point
Studer	Ileum 40–44 cm double-folded U + isoperistaltic afferent limb	Refluxing end-to-side	Simple, no staples, suits short ureters; stricture <3%; most popular
Hautmann	~70 cm ileum, W/M	Refluxing into W ends (since 1997; stenosis 9.5% → 1%)	Largest initial capacity → earlier continence, but more late retention and reabsorption
T pouch	44 cm U + 15 cm T limb in serosal trough (flap valve)	Antireflux	Accepts dilated/short ureters; more secondary procedures than Studer
Serous-lined tunnel (Abol-Enein)	40 cm ileum W	Extramural serosal trough	Durable antireflux; needs long non-dilated ureters
Hemi-Kock	61 cm ileum U + intussuscepted nipple	Antireflux nipple	Late nipple stenosis, extussusception, staple stones — abandoned
Camey II	65 cm transverse U (Barre: Z)	Le Duc	Detubularised successor of Camey I
Padua (VIP)	~42 cm ileum, spiral funnelled neck	Le Duc	Used robotically intracorporeal
Mainz III / right colon / sigmoid (Reddy)	Ileocaecal or colon	Submucosal tunnel	When ileum unusable; colonic pressure waves; sigmoid worst night continence

Key steps

ALGORITHM

Studer: isolate ileum 15–20 cm proximal to ileocaecal valve (distal division in avascular plane; short proximal mesenteric division) → discard ~5 cm proximally for mobility → 40–44 cm reservoir (two 20–22 cm limbs) + afferent limb (longer if ureters short) → open near mesentery → oversew back wall (running 3-0 polyglycolic acid) → fold opposite way, close → stented end-to-side ureteroileal anastomoses → urethral anastomosis at dependent point

- **T pouch:** distal 3–4 cm of the afferent limb laid in a serosal trough between the two limbs, fixed through mesenteric windows (Deaver) to keep its blood supply, covered by ileal flaps (flap valve). Tapering the limb was abandoned — late ostial stenosis.

- **Hautmann:** opened antimesenterically except a 5 cm U-flap at the urethral point; button excised for urethral anastomosis. Leaving W ends long accommodates short ureters; a 40 cm version limits retention and reabsorption.

> **KEY POINT** Hautmann vs Studer is a capacity trade-off: a bigger initial pouch is continent sooner but empties worse and absorbs more later.

6 · CONTINENT CUTANEOUS RESERVOIRS

The central problem is a durable, easily catheterisable continence mechanism. Nipple valves (Kock) are now historical.

Stoma site	Advantages
Umbilicus	Less stenosis (especially appendix); easiest for paraplegics; near-invisible — wear a medical alert bracelet
Lower quadrant (through rectus, below bikini line)	Concealed under clothing

Efferent mechanism	Principle	Notes
Appendix (Mitrofanoff) tunnelled in caecal taenia	Flap-valve compression in submucosal tunnel	Simplest, does not kink, excellent continence; needs ≥5–6 cm, ≥12 Fr; takes only 14–16 Fr catheters
Mitchell caecal lengthening	Stapled caecal tube in continuity with appendix	Adds length; less stenosis
Yang–Monti tube	2–3 cm ileum retubularised transversely	When appendix absent
Tapered ileum tunnelled into colon	Flap valve	Unlimited length
Indiana: imbricated ileocaecal valve + tapered ileum	Plicated valve + narrowed neourethra	Reliable; bowel frequency, vitamin B12 loss

- **Indiana pouch**: entire right colon + ~10 cm terminal ileum, Heineke–Mikulicz fold, ureterocolonic implants; valve double-imbricated with non-absorbable Lembert sutures (two rows ~8 mm apart) over 3–4 cm over a 16 Fr catheter; remaining ileum tapered with a stapler; neourethra to umbilicus or rectus. **Why the whole colon is opened:** a partly opened colon kept enough caecal peristalsis to overcome the valve; pressure profiles showed continence is confined to the valve, so imbrication is needed only there.
- **Right colon pouch** (authors' choice; Penn/Mitrofanoff-based): ~30 cm colon; intact terminal ileum as afferent limb (ureteroileal anastomosis; ileocaecal valve acts as antireflux); appendix opened to take ≥12 Fr, laid in a 4–5 cm taenial trough and covered (Mitrofanoff), matured at umbilicus; caecum fixed to posterior rectus fascia. Unsuitable appendix → Yang–Monti.
- Intra-op: fill pouch, check continence and easy catheterisation, fix reservoir to abdominal wall (prevents migration, kinking, false passage).

7 · PERIOPERATIVE CARE

- **Enhanced Recovery After Surgery (ERAS)**: counselling; prehabilitation and smoking cessation; no oral mechanical bowel preparation (authors still prepare when using colon), carbohydrate loading; restrictive fluids, normothermia; no nasogastric tube; early diet and mobilisation; opioid-sparing analgesia; **alvimopan** (μ-opioid antagonist) cut ileus from ~23% to ~7%. Ileus is the commonest gastrointestinal complication. Up to 70% are malnourished/frail → nutritional prehabilitation.
- **Drainage** (authors): 7 Fr single-J stents tied to a 24 Fr urethral (or caecostomy) catheter, irrigated routinely to clear mucus; drain out when low; review ~day 10; catheter out at **3 weeks**; no routine

pouchogram unless drain output is significant. Then frequent timed voids, lengthened progressively, plus pelvic floor exercises.
- Robotic intracorporeal diversion → *Ch 135*.

8 · RESULTS AND COMPLICATIONS

Early (<90 days) morbidity is not higher than with ileal conduit; continent cutaneous (colon) has more infection and dehiscence.

Complication	Key facts	Management
Urine leak	Commoner (long suture lines)	Catheter, stents; undrained → percutaneous drain ± bilateral nephrostomies; **avoid early open repair**
Bacteriuria	Very common; bowel cannot inhibit bacteria	Treat only if symptomatic or *Proteus/Pseudomonas*; febrile urinary tract infection (UTI) → look for obstruction, stones, residual; recurrent → nitrofurantoin/methenamine suppression
Ureteroileal stricture	Same as conduit; technique-dependent; can appear after 10 years, often silent	Suspect on CT/ultrasound; confirm by pouchogram, retrograde or antegrade study. Endoscopy (retreatment >50%) inferior to open repair
Pouch stones	Kock (staples) and metal-stapled pouches	Endoscopic
Pouch perforation	Rare; radiation, chronic retention; acute abdomen	Cystogram; catheter ± drains if stable
Pouch–vaginal fistula	5–10%; vaginal excision, radiation	Lateral cystogram before catheter removal; methylene blue if late leak. Will not heal with drainage after the first weeks → transvaginal → transabdominal repair or cutaneous conversion
Stomal stenosis (cutaneous)	Up to ~20%; commoner after 90 days and radiation	Usually open revision
Efferent limb necrosis	Rare, ischaemic	Flexible endoscopy for level; below fascia → explore
Ventral/incisional hernia	Common (straining to void, smoking); impairs emptying	Repair

Continence

- Daytime continence develops over **3–6 months**, ultimately **80–90%**. Night continence lags into year 2; persistent nocturnal incontinence **20–50%**. Risk: age >65, colon, diabetes, no nerve sparing, hysterectomy.
- **Why night leakage:** no sphincter–detrusor reflex, lower nocturnal sphincter tone, and bowel secretes water (loss of nocturnal concentration). Treat by emptying at bedtime and alarm-waking 2–3 times.
- Evaluate only after the 12–18-month plateau; exclude fistula. Pelvic floor physiotherapy with biofeedback; urodynamics to exclude pressure waves (colon pouches).

Stress incontinence	Men	Women
Bulking agent	Low leak pressure; marginal	Mild only; success <50%, poorly durable
Definitive	Artificial urinary sphincter or male sling	Pubovaginal sling — most then need catheterisation
Safety rule	No blind needle passage behind pubis (adherent pouch/bowel) → bone anchors or prepubic route	

- Cutaneous stomal leak is rare; persistent → surgical revision of the efferent mechanism; minor → bulking agent at the pouch entrance.

Retention and hypercontinence

- Presents as recurrent infection, overflow incontinence, palpable pouch or new hydronephrosis — rarely acute.
- Men 4–10% (rises to ~21% by 10 years); risk: excessive bowel length, prostate-sparing. **Women 20–30%**; up to ~60% self-catheterise on questionnaires.
- Female mechanism is **mechanical**: pouch falls back into the wide pelvis, angulating the pouch–urethral junction (neocystocele). Prevention: omentum behind pouch, vaginal suspension, sacrocolpopexy, uterine preservation.
- Treatment: clean intermittent self-catheterisation (drugs do not help); exclude stricture (true anastomotic stricture ~1%) or recurrence with rectal/vaginal examination and cystoscopy. An obstructing mucosal flap can be incised endoscopically.

> **KEY POINT** Men fail by **leaking** (sphincter without its reflexes) — augment the sphincter. Women fail by **not emptying** (unsupported pouch kinks) — support the pouch.

9 · FOLLOW-UP

- Phases: first 4 months → ureteroileal stricture; 4 months–3 years → cancer recurrence; >3 years → stones, late obstruction, urethral/upper-tract tumour.
- Every 4 months in year 1, 6-monthly to 3 years, then annually: examination (pelvic), metabolic panel, blood count. computed tomography (CT) abdomen–pelvis + chest imaging, intensity by stage, to 5 years.
- Annual: vitamin B12; urethral wash cytology in men (especially prostatic CIS); bimanual examination and voided cytology in women (urethral recurrence is invisible on CT). Cystoscopy for new retention, altered voiding or haematuria.
- After 5 years, lifelong: upper-tract ultrasound, metabolic panel, blood count — late complications are silent.
- Quality of life: no diversion proven superior; neobladder better body image, conduit better urinary function. Informed choice reduces regret.

10 · LANDMARK TRIALS

Trial	Design	Result	Take-home
USC Studer vs T pouch	Randomised, 484 RC patients	No difference in renal function or infection at 3 years; T pouch more secondary procedures	Antireflux adds cost without benefit
RAZOR	Open vs robotic RC, extracorporeal diversion	Continent diversion only 21%	Low uptake even with expert surgeons
iROC	Open vs robotic RC, intracorporeal diversion	Continent diversion 11%	Intracorporeal approach further reduces continent diversion

11 · MCQ ONE-LINERS

1. Mature neobladder target capacity: ≥500 mL at low pressure.
2. Detubularisation abolishes coordinated peristaltic contractions; a sphere maximises volume per surface area.
3. Most distensible bowel muscle: ileal circular; least: colonic longitudinal.
4. Renal threshold for continent diversion: eGFR >35–40 mL/min or creatinine <1.7–2.2 mg/dL.
5. Strongest male urethral recurrence risk: prostatic stromal invasion.
6. Strongest female urethral risk: bladder-neck tumour; no skip lesions to urethra.
7. Bladder CIS does not predict urethral recurrence.

8. Best test for urethral involvement: intraoperative frozen section of urethral margin.
9. Orthotopic neobladder protects against urethral recurrence (HR 0.44).
0. Commonest technical reason to abandon orthotopic diversion: short/fatty mesentery.
1. Self-catheterisation after neobladder: 10–20% of men, ~30% of women.
2. Rhabdosphincter lies anterolateral to urethra, supplied by pudendal nerve.
3. Omental flap between vagina and pouch prevents pouch–vaginal fistula.
4. Hautmann W uses ~70 cm ileum; Studer 40–44 cm plus afferent limb.
5. Kock nipple late problems: stenosis, extussusception, staple stones.
6. Lowest ureteric stricture: direct Wallace/Leadbetter refluxing anastomosis, ~3–6%.
7. Studer vs T pouch randomised controlled trial (RCT): no renal difference; T pouch more reoperations.
8. Mitrofanoff appendix channel needs ≥5–6 cm length and ≥12 Fr calibre.
9. Indiana pouch continence: imbricated ileocaecal valve plus tapered terminal ileum.
20. Undrained urine leak: percutaneous drainage/nephrostomy, not early open repair.
21. Alvimopan is a peripheral μ-opioid antagonist reducing postoperative ileus.
22. Night-time incontinence persists in 20–50%; bowel water secretion contributes.
23. Retention: women 20–30% vs men 4–10%; female cause is mechanical angulation.
24. Treat retention with intermittent self-catheterisation; drugs are ineffective.
25. Treat bacteriuria only if symptomatic or Proteus/Pseudomonas.
26. Persistent cutaneous stomal incontinence requires surgical revision of efferent mechanism.
27. Female urethral recurrence essentially never occurs without bladder-neck involvement.
28. Indiana valve imbricated over a 16 Fr catheter with non-absorbable Lembert sutures.
29. Prior bowel resection: reuse the old anastomosis as one reservoir end.
30. Pouch–vaginal fistula beyond first weeks needs surgical repair, not drainage alone.
31. Annual vitamin B12 and lifelong upper-tract imaging after continent diversion.

135 Minimally Invasive Urinary Diversion

Reading time ≈ 27 min (slow pace)

1 · CONTEXT

- Robot-assisted radical cystectomy (RARC) is oncologically equivalent to open in randomised controlled trials (RCTs, e.g. RAZOR) — but those trials used **extracorporeal** diversion (ECUD); intracorporeal urinary diversion (ICUD) was not tested.
- International Robotic Cystectomy Consortium (IRCC): ICUD rose ≈**11%/year**, driven by **ileal conduit** (commonest robotic diversion).
- Decline of continent reservoirs predates robotics: no high-quality evidence that continent diversion gives better quality of life (QoL). In iROC only ≈11% of diversions were continent.

2 · PATIENT SELECTION AND CONTRAINDICATIONS

High body mass index, older age, locally advanced disease and prior abdominal surgery are no longer exclusions. Three separate decisions — a contraindication to one is not a contraindication to all:

Decision	Contraindication
Minimally invasive route	Poor cardiopulmonary compliance → cannot tolerate steep Trendelenburg for a long ICUD (relative; eased by da Vinci Xi and AirSeal, allowing flatter position)
Use of ileum	Short bowel; severe inflammatory bowel disease; radiation enteritis
Neobladder (absolute)	Urethral involvement distal to prostate; creatinine **>2 mg/dL**; poor hepatic function — same as open

ALGORITHM

Pasadena Consensus learning pathway: RARC + extracorporeal diversion → intracorporeal conduit (≈30–100 cases) → continent diversion (≈25–50 cases). Target time: **≤4 h conduit, ≤5 h neobladder**.

3 · PREOPERATIVE CARE AND PERIOPERATIVE BUNDLE

Enhanced recovery after surgery (ERAS; >20 elements) removes what delays gut recovery: alvimopan, goal-directed fluids, immediate oral diet, opioid-sparing analgesia (high-dose paracetamol/ketorolac rather than epidural or opioid patient-controlled analgesia). Result: less ileus, shorter stay, fewer complications/readmissions. Caution: stay **<5 days** ≈ doubled 90-day major complications — early discharge shifts complications home.

Element	Recommendation (reason)
Fasting	Solids to 6 h, clear fluids to 2 h
Bowel preparation	**None**, no oral antibiotics (electrolyte upset in elderly; liquid spillage on intracorporeal bowel opening; no benefit)
Thromboprophylaxis	Pneumatic compression + stockings; weight-based low-molecular-weight heparin before and up to **4 weeks** after
Antibiotics	Broad-spectrum intravenous within **1 h**

4 · POSITION AND PORTS

- Lithotomy, steep Trendelenburg (Xi integrated table permits flatter position without undocking); Hasson entry if previous surgery.
- **6–7 transperitoneal ports placed more cephalad** — reach over small bowel and extended node dissection to aortic bifurcation.
- Robotic arm can alternate with assistant instrument inside a 15-mm port (stapling).

- Robotic arms can be undocked and Trendelenburg reduced for the urethra–neobladder anastomosis.
- Zero-degree lens throughout (30° down for deep female pelvis).

Port	Size	Position / use
Camera	12 mm (8 mm Xi)	Above umbilicus, midline or left
Right/left arms	8 mm	Umbilical level, lateral to rectus, symmetrical
Fourth arm / hybrid	8 or 12 mm	Above and medial to anterior superior iliac spine
Extra left (neobladder)	12–15 mm	Above/medial to left 8-mm port; stapling
Assistants	5–15 mm; 12 mm left upper quadrant	Between camera and right arm
Suprapubic	12 mm (stapler) or 5 mm (stents)	Bowel reanastomosis; extended to Pfannenstiel for specimen

5 · ANATOMIC CONSIDERATIONS (NERVE SPARING)

The same autonomic plexus serves sexual function and neobladder continence — nerve sparing is a continence operation too. Select for anatomically safe tumour, low burden and adequate baseline sexual function. Seminal vesicle dissection is athermal after the vasal ampullae; the middle/lower seminal vesicle may be left. Full lateral (veil of Aphrodite) preservation needs apical dissection after the bladder is dropped.

	Male	Female
Danger points	(1) presacral/medial common iliac (node dissection) → (2) high bladder pedicle ligation → (3) seminal vesicles/Denonvilliers fascia → (4) prostatic pedicles; apex	Lateral vaginal wall and bladder base (inferior hypogastric plexus most vulnerable)
Manoeuvre	Athermal, intrafascial, traction-free pedicle control; meticulous periurethral apical dissection	Preserve tissue around proximal urethra/lateral to bladder neck, anterior vaginal wall; extract via small Pfannenstiel, not transvaginally
Sphincter	One-third overlapped by dorsal venous complex; dorsal penile nerve branches within **3 mm** of apex	Mainly mid–distal third; continence survives loss of bladder neck and cranial third

6 · INTRACORPOREAL ILEAL CONDUIT

ALGORITHM
Left ureter under sigmoid mesentery to right → 12 cm ileum, 15–20 cm from ileocaecal valve → marionette stitch → isolate with 45-mm stapler → ureteroileal anastomoses (left first) + stents → bowel reanastomosis → stoma.

- **Marionette stitch**: 1-0 silk on Keith needle through hypogastrium and distal (stoma) end, held **untied** — replaces the missing hand: moves/lowers the conduit into the robot's optimal working range.
- Avoid redundant segment (stasis, infection, renal loss) → *Ch 133*; leave enough terminal ileum. Duplex ureters: implant separately or together by calibre. Short left ureter: retract sigmoid with fourth arm.
- Isolation: two mesenteric windows → stapler proximally and distally → cut ends tagged with silk (prevents malrotation); avoid narrowing the conduit base (mesenteric supply). Washout abandoned (no benefit, spillage).
- **Enterotomy**: athermal cold-scissors buttonhole with serosal removal — adopted after thermal-hook slit enterotomy gave more strictures. Retroperitonealise anastomosis with peritoneal flap.

- Stenting after posterior wall: suction tube via conduit → 8.5-Fr single-J over guidewire; **guidewire removed only after anchoring** (no tactile feedback). Stent ends out via 15-mm port, not clamped to drapes (dislodgement).
- First suture conduit outside-in, ureteric angle inside-out, conduit bite perpendicular to staple line → no malrotation or back-walling. Distal ureter sent for histology; no routine frozen section.
- Bowel restored after both ureters (easier conduit handling); robot-sewn ileoileal anastomosis feasible.
- Stoma: cruciate fascial incision with stay sutures → clamp through stoma site grasps marionette and stents → Jackson-Pratt drain → undock → conduit delivered; check orientation once pneumoperitoneum is lowered.

Bricker	Wallace
Each ureter end-to-side separately (refluxing)	Ureters joined as "Y", single anastomosis to proximal conduit
Easy; renal units kept separate	One bowel anastomosis

7 · INTRACORPOREAL NEOBLADDER

Key difference from open: **early neobladder–urethral anastomosis** (Karolinska) — anchor the most dependent ileum to the urethra first, then build the pouch in situ. Undock and reduce Trendelenburg to 10–15° (unneeded with Xi). Design principles → *Ch 134*.

Design	Ileum	Signature
Karolinska (modified Studer)	50 cm	10-cm afferent limb; Penrose sling; Bricker or Wallace
USC (Studer)	65 cm (longest)	44+16 chimney+5 discarded; 90° anticlockwise rotation
UCLH Pyramid	50 cm	Koie-derived cross-fold; suprapubic tube
Saint-Augustin modified Y	40 cm (shortest)	Refluxing, uncrossed ureters
Padua	42 cm	Only stapled pouch; split-nipple antireflux
FloRIN	50 cm	Asymmetrical U; L-shaped neo-trigone; antireflux
Roswell Park (Hautmann W)	45–50 cm	Stays A–D; one trough at a time; chimneys

Key steps (Karolinska and Roswell Park)

ALGORITHM
Urethral anastomosis (Van Velthoven barbed suture) → detubularise distal 40 cm (cold scissors, over 24-Fr chest tube) → posterior plate → ureters to afferent limb/chimney, stented → close anterior wall after 22-Fr catheter confirmed.

- Roswell Park W: one trough at a time → less spillage, orientation kept; Foley tip tagged to stays acts as marionette; end-to-side ureters into chimneys.
- Single-J preferred: long enough to fix; chromic stitch to pouch and a second to Foley for removal. Foley used for irrigation.
- Bowel restored via suprapubic port: 60-mm side-to-side stapled anastomosis; close mesenteric window.

Manoeuvres for tension-free urethral anastomosis

One of the hardest steps: stretch must be judged without tactile feedback, often in steep Trendelenburg.

- Preserve maximal urethral length (open dorsal venous complex without initial suturing).
- Mobilise sigmoid out of pelvis; reduce Trendelenburg; perineal pressure; Foley guides needle.
- Barbed suture; detubularise and stretch bowel; Penrose drains for traction (Karolinska); Rocco-type stitch (USC).
- Incising mesenteric peritoneum adds length but risks pouch ischaemia.

Intracorporeal continent cutaneous diversion (Indiana pouch)

ALGORITHM
Mobilise right colon to mid-transverse → divide ileum 15 cm from ileocaecal junction and transverse colon proximal to middle colic → restore ileocolic continuity immediately (prevents malrotation) → resect appendix base → detubularise along anterior taenia, fold vertically → ureters through posterior plate, double-J; 24-Fr caecostomy via appendiceal stump → taper ileum over 14-Fr catheter, imbricate ileocaecal valve → umbilical stoma.

KEY POINT Capacity comes from the detubularised right colon; continence from the tapered ileum and imbricated ileocaecal valve.

8 · POSTOPERATIVE CARE: DRAINS, CATHETERS, STENTS

Avoid overhydration (elderly); no routine nasogastric tube (Cochrane). Stenting and drainage aid bowel recovery and reduce metabolic acidosis. Irrigate neobladder **8-hourly** (mucus plugs).

Tube	Removal
Urethral Foley as pelvic drain (conduit)	Day 1
Blake drain (neobladder)	When output falls
Neobladder catheter	**3–4 weeks**, after leak-free pouchogram
Externalised stents / internal double-J	≈1 week / cystoscopically 3–4 weeks
Feeding-tube stents tied to Foley	With catheter (avoids cystoscopy; lost if Foley falls out)

9 · OUTCOMES

- **Operative time** is a quality measure (prolonged time → more complications, mortality). Strongest determinant: diversion type (conduit ≈70 min shorter). ICUD ≈ ECUD time (redocking cancels saving). Proficiency ≥**66 RARCs**.
- **Complications**: mostly diversion-related, infection commonest; two-thirds of reoperations after 3 months → 90-day reporting underestimates. Meta-analysis of RCTs: no difference in 90-day major complications, open vs robotic.
- **Parastomal hernia**: ≈one-third by 3 years; risks — long operation, fascial defect ≥30 mm, low estimated glomerular filtration rate (poor healing).; robotic mesh hernioplasty feasible and durable.
- **ICUD vs ECUD (IRCC)**: early advantages (fewer gastrointestinal complications, readmissions) confounded by selection; after adjustment, approach is neutral for high-grade complications and readmission; transfusion lower.
- **Reoperation**: cumulative ≈half by 5 years; predictors — neoadjuvant chemotherapy and any high-grade complication.
- **Uretero-enteric stricture**: rises over 5 years; prevent with athermal enterotomy and retroperitonealisation. Robotic revision equals open. Indocyanine green (ICG) localises ureter and stricture.

Function and urodynamics

KEY POINT Why is night continence worse? Daytime continence relies on the urethrosphincteric guarding reflex (urine in posterior urethra → sphincter contraction). In deep sleep this reflex is inactive, so continence depends on reservoir capacity — hence detubularisation, nerve and sphincter preservation, night alarm and limiting nocturnal output. Assess at 6–12 months, when capacity is maximal.

- Incontinence factors: age, reservoir volume/shape, functional urethral length, nerve supply. Good continence predicted by good Eastern Cooperative Oncology Group (ECOG) status and retained filling sensation. Up to 25% of men need clean intermittent catheterisation (CIC) after open neobladder.

- Intracorporeal series: day continence 68–100%, night 55–85%, CIC up to 11%.
- Urodynamics: capacity generally adequate, but emptying (high residual, low flow) and night continence lag; only FloRIN matched its open comparator (higher compliance).
- QoL: physical domains dip to 6 months, recover by 24 months except sexual function.

10 · LANDMARK TRIALS

Trial	Design	Result	Take-home
RAZOR	RCT RARC vs open, extracorporeal diversion	Oncological equivalence	Says nothing about ICUD
iROC	Multicentre RCT, totally intracorporeal vs open	Open: worse QoL at 5 weeks, more disability at 5 and 12 weeks; no difference after 12 weeks; 11% continent	ICUD gains are early and recovery-related

11 · MCQ ONE-LINERS

1. Commonest robotic diversion: ileal conduit.
2. Annual rise in ICUD (IRCC): ≈11%.
3. Creatinine contraindicating neobladder: >2 mg/dL.
4. Main contraindication to robotic route: poor cardiopulmonary tolerance of steep Trendelenburg.
5. Bowel preparation before ICUD: none, no oral antibiotics.
6. Low-molecular-weight heparin duration: up to 4 weeks.
7. Stay <5 days: ≈twofold 90-day major complications.
8. Target time: ≤4 h conduit, ≤5 h neobladder.
9. Marionette stitch: untied, held by instrument.
0. Conduit: 12 cm ileum, 15–20 cm from ileocaecal valve.
1. Stricture-reducing enterotomy: athermal buttonhole with serosal removal.
2. Guidewire removed only after stent anchoring.
3. Karolinska innovation: early neobladder–urethral anastomosis.
14. Only stapled intracorporeal pouch: Padua.
15. Longest ileum: USC (65 cm); shortest: Saint-Augustin Y (40 cm).
16. Neobladder catheter out: 3–4 weeks after pouchogram.
17. Strongest determinant of operative time: diversion type.
18. IRCC proficiency: ≥66 RARCs.
19. Parastomal hernia: ≈one-third by 3 years.
20. Most reoperations: diversion-related, after 90 days.
21. Night incontinence: guarding reflex inactive in sleep.
22. iROC: QoL benefit lost after 12 weeks.
23. Indiana pouch continence: tapered ileum + imbricated ileocaecal valve.
24. Reoperation predictors: neoadjuvant chemotherapy, high-grade complication.
25. ICG aids robotic uretero-enteric stricture revision.
26. Good continence predictors: ECOG status, retained filling sensation.

124 Bladder Surgery for Benign Disease

Reading time ≈ 70 min (slow pace)

1 · SURGICAL ANATOMY ESSENTIALS

- Extraperitoneal hollow organ: mucosa, lamina propria, detrusor, perivesical fat. Derived from the urogenital sinus (endoderm).
- **Anterior**: pubis and retropubic space (space of Retzius), the usual access, also used for prostate and proximal urethral surgery; transversalis fascia separates bladder from the abdominal wall muscles. **Posterior**: peritoneum; peritoneal reflection near the prostatorectal junction (men) or vesicouterine space (women) — incising it opens the space behind bladder and trigone. Denonvilliers fascia separates bladder/prostate from rectum; vagina lies behind the bladder in women. **Lateral**: pelvic sidewall, obturator space; vas or round ligament.
- **Medial umbilical ligaments** = obliterated umbilical arteries; **median umbilical ligament** = urachus (allantois remnant; failure → patent urachus, sinus or cyst).
- Ureter crossings: **uterine artery** over ureter near the internal os ("water under the bridge"); **vas** over ureter near the seminal vesicle tip. Distal ureter runs medial to the medial umbilical ligament to the trigone.
- Superior vesical (from umbilical artery) and inferior vesical (from internal iliac; vaginal artery in women) arteries form posterolateral pedicles; dividing one aids mobilisation. Venous plexus → internal iliac veins. Lymph: mainly external iliac, also obturator, internal and common iliac.
- Nerves: sympathetic T10–L2, parasympathetic S2–S4; pudendal nerve (Onuf's nucleus, S2–S4) to the urethral sphincter.

> **KEY POINT** Three landmarks organise every benign bladder operation: the space of Retzius for access, the peritoneal reflection for the posterior bladder, and the distal ureter for safety — knowing where each lies decides whether the operation can stay extraperitoneal.

> **KEY POINT** Common aftercare for almost every operation here: watertight **two-layer 2-0 polyglactin** closure (mucosa first, then seromuscular), leak test by filling, large-bore Foley ≈ **1 week**, pelvic drain on self-suction out at ≈ **48 h** when output is low, **cystogram before catheter removal**. High drain output → compare drain-fluid creatinine with serum creatinine (raised = urine leak).

2 · BLADDER DIVERTICULA

A diverticulum is an outpouching of mucosa and lamina propria **without functional muscularis propria**; its scattered, disorganised detrusor fibres cannot contract in a coordinated way, so it empties poorly. Stasis causes urinary tract infection (UTI), stones and a sensation of incomplete emptying; a large one may compress the ureter near its insertion and cause hydronephrosis. Cancer risk within diverticula is increased. A fibrous **pseudocapsule** forms around it and provides the surgical cleavage plane.

Feature	Acquired	Congenital
Who	Adults, male predominance	Children
Cause	High pressure: bladder outlet obstruction (BOO; benign prostatic hyperplasia [BPH] commonest, stricture, prolapse) or non-compliant neurogenic bladder	Intrinsic detrusor weakness
Number / site	Often multiple, variable size; with trabeculation	Solitary, posterolateral beside ureteric orifice (**Hutch**)
Association	Stones, UTI, cancer risk	Vesicoureteral reflux (VUR) generally present (detrusor deficient at hiatus)

Feature	Acquired	Congenital
Implication	Treat outlet/compliance before or with diverticulum	Plan reimplantation; still screen for voiding dysfunction

Other adult causes: posterior urethral valves, prune-belly syndrome, Ehlers-Danlos and Williams syndromes, iatrogenic (women after anti-incontinence surgery). Most acquired diverticula are asymptomatic and incidental.

Work-up

- **Cystoscopy**: enter and inspect every diverticulum (cancer would change treatment entirely); note neck size, number, volume, distance from ureteric orifice, trabeculation, intravesical prostatic lobe, stricture or bladder neck contracture (each treated before or with the diverticulum). Coexisting stones removed at diverticulectomy.
- **Voiding cystourethrogram (VCUG)**: number, size, VUR, relation to ureter → decides preoperative stent (identification, protection) vs simultaneous reimplantation; shows diverticula hidden among trabeculae and bladder neck obstruction (hypertrophy, dyssynergia, valves, stricture).
- Computed tomography (CT) urogram (triphasic with delayed excretory phase): hydronephrosis (compression or VUR through a weak tunnel), ureteric course and lateral displacement, stones. Ultrasound shows hydronephrosis but not ureter or stones well. No contrast possible → ultrasound/non-contrast CT plus intraoperative retrograde pyelography and cystography.
- Cytology in all; flow rate or urodynamics (contractility, voiding pressure); prostate volume (transrectal ultrasound, CT, MRI or cystoscopy) if treating BPH; **negative urine culture** before surgery.

Situation	Management
Small, asymptomatic, uncomplicated	Observe, or endoscopic fulguration/resection
Large with residual, recurrent UTI, stones, pain	Diverticulectomy
VUR or ureteric obstruction	Diverticulectomy ± reimplantation
Cancer in diverticulum	Partial or radical cystectomy — not diverticulectomy

Operative principles

- Treat the cause at the same sitting: transurethral prostate surgery, or simple prostatectomy for glands **>80 cc**; augmentation for neurogenic bladder. **Endoscopic outlet surgery goes first** — irrigation afterwards would distend the bladder and disrupt fresh suture lines.
- Localise the sac: Councill-tip catheter over a guidewire into the diverticulum for selective filling (separate Foley fills the bladder), or flexible cystoscopic transillumination (best when multiple). Ureteral catheter if close to ureter.
- Open surgery is extraperitoneal, even for posterior diverticula (less visceral injury and intraperitoneal urine leak, a major source of morbidity); laparoscopic (LBD) and robotic (RBD) diverticulectomy are usually transperitoneal (wide working space). Open remains the gold standard; complex (large, posterior) or multiple diverticula needing reimplantation traditionally open, though minimally invasive is reasonable in experienced hands.

Open intravesical technique

ALGORITHM

Supine, anterior superior iliac spine (ASIS) at table break, slight flexion → 18 Fr Foley, fill ≈ 200 mL, plug → lower midline incision, split recti, incise transversalis fascia (distension pushes the peritoneal reflection up) → sponge-stick develops Retzius and gutters, sweeping peritoneum off the dome → vertical midline cystotomy dome to above bladder neck, 2-0 stay sutures at apices → inspect mucosa, identify orifices (5 Fr ureteral catheter if concern) → excise sac → close defect → 20–22 Fr Foley, two-layer cystotomy closure, leak test → optional suprapubic catheter through a separate punch cystotomy → pelvic drain.

Open	Intravesical	Combined intra-/extravesical
Delivery	Allis clamp through the neck everts sac into bladder; or (not evertible) sac dissected out of pseudocapsule from inside with Allis countertraction on the neck; mucosal incision ≈ **0.5 cm** round neck	Same mucosal incision; finger or suction tip in sac pushes it out; Kittner clears fat; extravesical neck incised circumferentially
Hazard	Ureteric orifice injury	**Excess bladder wall resection** if outer and inner incisions do not meet (new plane)
Closure	Two layers 2-0 polyglactin with secure seromuscular layer (prevents recurrence); no drain in remnant capsule	

Laparoscopic / robotic technique

- Lithotomy, steep Trendelenburg; cystoscopy first (retrograde pyelogram, selective catheterisation or transillumination). Pneumoperitoneum **15 mm Hg**; robotic assistant port contralateral to the diverticulum (iliac bone restricts an ipsilateral port).
- Incise peritoneum over the sac; for large/multiple sacs drop the bladder (incise lateral to both obliterated umbilical ligaments, divide urachus) → divide adhesions to the neck → circumscribe the neck from anteriorly with traction on the sac → excise completely, specimen in bag. Keep the ureter in view; it may need dissecting off the sac.
- Transvesical variant (open bladder in midline, invert sac, circumscribe mucosa) — use cautiously if ureter involved. Stapler excision risks stones on staples.
- If the diverticulum involves the ureteral hiatus: amputate the neck, separate the ureter, close the opening, create a new anterior-wall hiatus, double-J stent, reimplant.
- Close in two layers 2-0 polyglactin (bladder hitched anteriorly to expose the defect); fill to test; peritoneum may be closed to separate compartments; drain via trocar site.
- Alternatives: extraperitoneal laparoscopy (balloon-dilated Retzius space), transvesical pneumovesicoscopic surgery, laparoendoscopic single-site surgery (LESS) and single-port robot — useful for posterior diverticula.
- Aftercare: Foley ≈1 week, drain out at 24–48 h if low output, cystogram before removal; periprocedural antibiotics optional.

Complications and their management

- **Urinary extravasation commonest** (similar open and minimally invasive): high drain output, or pain and fever from infected urinoma; after transperitoneal surgery ileus and electrolyte disturbance (urinary ascites). Confirm with drain creatinine; define with CT cystography.

ALGORITHM
Asymptomatic leak → keep Foley and drain → symptomatic collection → percutaneous drain → drain study and cystogram before removal → persistent → nephrostomy tubes vs exploration and closure.

- Unrecognised ureteric injury: acute kidney injury (AKI), flank pain, extravasation features → excretory or retrograde imaging (± drain urinoma) → retrograde stent first; if it fails, nephrostomy and delayed repair.
- Others: UTI, fistula, bladder spasm, wound infection, bowel injury.

> **KEY POINT** Diverticulectomy is a dissection problem, not a reconstruction problem: find the sac, stay in the sac–pseudocapsule plane, keep the ureter in view, close the seromuscular defect securely — and fix the outlet first.

3 · BLADDER STONES, FOREIGN BODIES AND FISTULAE

- Cause is **stasis**: BOO (BPH), neurogenic bladder; also augmented/substituted bladders (stasis, bacteria, mucus, metabolic change) and foreign bodies as nidus (usually iatrogenic: mesh, migrated intrauterine device, needles, wires).

- Most asymptomatic; otherwise haematuria, irritative voiding, recurrent UTI, incontinence or retention.
- **Non-contrast CT** best measures size/volume (guides choice) and shows prostate volume or hydronephrosis; cystoscopy only if haematuria, suspected cancer or prostate assessment (lobes, prostatic urethral length) needed; urodynamics if neurogenic or for BPH planning.
- Treat the stone **and** the cause; the concomitant operation often chooses the route: very large prostate → transvesical simple prostatectomy with cystolithotomy; large diverticulum → diverticulectomy; refractory non-compliant neurogenic bladder → enterocystoplasty with cystolithotomy.

Approach	Best for	Advantage	Limitation
Transurethral cystolitholapaxy	Most stones	No incision	Prolonged instrumentation → urethral stricture; inefficient for big burdens
Percutaneous cystolithotripsy	Large stones	Small incision, nephroscope (larger channel, efficient fragmentation), short catheter time and stay, no stricture risk	Contraindicated if prior surgery, diversion or augmentation — bowel injury
Open cystolithotomy	Very large burden; with transvesical prostatectomy, diverticulectomy, augmentation	Historic gold standard; clearance, speed	Incision, longer catheter
Laparoscopic/robotic (transperitoneal or transvesical)	Large burden with other procedures; foreign bodies, eroded mesh	Minimally invasive	Technical demand

- **Open**: Pfannenstiel or midline (depends on concomitant surgery) → vertical cystotomy → remove stones, inspect for tumour and intravesical lobe, do prostatectomy/diverticulectomy → two-layer closure (mucosa, then imbricating layer) → drain.
- **Percutaneous**: distended bladder → 1-cm incision two fingerbreadths above symphysis → 18-gauge needle under ultrasound or cystoscopic vision (prior surgery: imaging to exclude interposed bowel) → stiff guidewire coiled in bladder → dilate to **30 Fr** sheath → 24–26 Fr nephroscope; stones <1 cm grasped intact, larger fragmented (pneumatic/ultrasonic); entrapment-bag option. Suprapubic catheter over the wire, clamped at 24–48 h for trial of voiding (or urethral catheter out at 24–48 h).
- **Transvesical laparoscopy**: trocars into bladder under cystoscopic vision, pneumovesicum **8–12 mm Hg**; avoids the peritoneum; suited to eroded mesh.
- Laparoscopic/robotic transperitoneal cystolithotomy: drop bladder → anterior cystotomy → remove stone or excise foreign body → two-layer closure → drain. Allows simultaneous minimally invasive prostatectomy or diverticulectomy.
- Aftercare: Foley ≈1 week (open/laparoscopic), cystogram; drain creatinine only if output high.
- **Fistulae** (colovesical, vesicovaginal) failing conservative care are an indication for partial cystectomy; refractory urinary fistula may justify simple cystectomy; single-port robotic transvesical vesicovaginal fistula repair is described.

4 · PARTIAL CYSTECTOMY FOR BENIGN DISEASE AND URACHAL SURGERY

- Benign indications: leiomyoma, paraganglioma (large, infiltrating or symptomatic benign tumours; most benign tumours need only removal of the stimulus or transurethral resection), endometriosis, urachal anomalies, fistulae.

- Selection is less strict than for cancer (multiplicity, carcinoma in situ [CIS] matter less). Main question: will the capacity loss harm quality of life? **Avoid with small capacity or severe dysfunctional voiding.** Exclude malignancy (excretory CT, cytology, cystoscopy, biopsy) because cancer requires wide margins and lymphadenectomy. Base lesions are harder; lesions near the orifice need reimplantation.

Lesion	Clue	Investigation	Treatment / trap
Leiomyoma	Women of childbearing age; obstruction, irritative symptoms, pain, haematuria; endovesical, intramural or extravesical	Transvaginal ultrasound, magnetic resonance imaging (MRI)	Small endovesical → transurethral resection (TUR); large/intramural → wide local resection (regrowth)
Paraganglioma	Episodic hypertension, palpitations, sweating; **hypotension/syncope on micturition or sex**; or painless haematuria, wall thickening	Plasma metanephrines; metaiodobenzylguanidine (MIBG) scan; yellow raised mucosa (intradetrusor)	**α-blockade before any instrumentation** (catecholamine surge, worst if asymptomatic); partial cystectomy preferred over TUR (recurrence)
Endometriosis	Cyclical haematuria/symptoms with menses; primary or iatrogenic	Ultrasound, MRI (T1 bright); cystoscopy limited (bluish lesion only if mucosal)	Partial cystectomy (disease extravesical)
Urachal anomaly (cyst, sinus, patent urachus, vesicourachal diverticulum)	Umbilical discharge, mucinuria, haematuria, infected mass, stone; risk of malignant change	Ultrasound/CT; VCUG or sinogram; cystoscopy of dome	Umbilical swab + urine culture → antibiotics ± incision and drainage; recurrent → excise whole urachus + dome cuff

Technique

- Route: dome/anterior lesions → extraperitoneal; posterior → transperitoneal (ureteric mobilisation from the iliac bifurcation; develop pubocervical fascia in women or Denonvilliers fascia in men for base lesions). Pelvic lymph node dissection unnecessary.
- Localise by intraoperative cystoscopy (intravesical) or ultrasound (intramural/extravesical); stay sutures away from lesion; **aspirate before opening to avoid spillage** (benign lesions also recur locally).
- Full-thickness excision preserving overlying perivesical fat, with ≈ **2-cm margin** if capacity allows; biopsies/frozen section of margins; reimplant if orifice involved; 22 Fr Foley, two-layer closure, drain.
- Minimally invasive: bedside assistant's flexible cystoscopy transilluminates the lesion; laparoscopic ultrasound for intramural lesions; outline the margin on the serosa first; ureteral catheter protects the ureter.
- **Urachal excision**: whole tract with peritoneal wings plus small dome cuff (prevents recurrence and malignant change); **no wide margin or frozen section** for benign disease; umbilicus may be kept if no malignancy and subcutaneous fluid drained. Open: periumbilical midline incision → circumscribe umbilicus down to peritoneum → enter peritoneum above umbilicus → with traction, excise urachus with peritoneal wings down to dome → distend bladder, transmural circumferential cuff → two-layer closure, Foley, drain. Minimally invasive: camera 5 cm above and 2 cm left of umbilicus; bladder left on the abdominal wall and urachus dissected last.
- Complications: extravasation commonest, then ureteric injury (managed as for diverticulectomy); reduced capacity with irritative symptoms — hence counselling.

> **KEY POINT** Why margins for benign lesions? Leiomyoma, paraganglioma and endometriosis are infiltrative and recur after incomplete excision — so benign partial cystectomy borrows cancer discipline (exclude cancer, no spillage, margin). Urachal surgery is the exception: the whole tract matters, a wide margin does not.

5 · SIMPLE (SUPRATRIGONAL) CYSTECTOMY

- Last resort, because irreversible and morbid, for end-stage refractory dysfunction: radiation cystitis, neurogenic bladder (spinal cord injury, multiple sclerosis), refractory incontinence, fistula, bladder neck contracture, refractory vesicourethral anastomotic stricture, interstitial cystitis/bladder pain syndrome (IC/BPS). Aim: preserve renal function or improve quality of life.
- Work-up: history, symptom scores, postvoid residual, urodynamics, cystoscopy, imaging. Must prove the bladder is the source: pain not truly vesical persists after cystectomy. IC/BPS success predictors: **small fibrotic bladder, small capacity under anaesthesia, Hunner lesions**.
- Diversion alone (without cystectomy) is an option in selected patients, but risks pyocystis, bleeding, pain and secondary urothelial cancer; many IC/BPS patients later need completion cystectomy.

Feature	Simple (benign)	Radical (cancer)
Prostate / anterior vagina	Preserved unless indicated	Removed
Lymphadenectomy	None	Performed
Bladder opened	Often — to find trigone and bladder neck	Avoided
Excision	Bivalve, amputate near trigone; stump oversewn or covered with peritoneum; residual mucosa cauterised	En bloc

Steps: ureters divided for diversion → bladder filled, bivalved in midline (anteriorly to bladder neck, posteriorly to cul-de-sac) → pedicles divided, lateral halves amputated → posterior peritoneum covers the defect → usually ileal conduit. Friable tissue may be removed piecemeal robotically. Care mirrors radical cystectomy; drains 5–7 days (out when drain creatinine equals serum); complications common (wound infection, UTI, transfusion, ileus); robotic surgery may reduce them.

6 · URETERAL REIMPLANTATION IN ADULTS

- Indications: distal ureteral trauma (open repair), intraoperative iatrogenic injury (same approach as index operation), **stricture** and delayed-presentation injury (commonest elective); with diverticulectomy or partial cystectomy. Primary VUR is rarely the indication in adults.
- Acute: inspection defines injury; after thermal or gunshot injury, debride even normal-looking ureter; partial injury → on-table retrograde pyelogram or proximal methylene blue.
- Elective work-up measures the gap: antegrade (CT urogram/nephrostogram) plus retrograde imaging; multiple segments lengthen the defect. Stricture causes: impacted stone, radiation, chronic inflammation. Exclude malignancy (cytology, ureteroscopic biopsy in smokers/elderly, frozen section of the margin). Atrophic kidney → renogram (decompress first if severely obstructed); **poorly functioning kidney → nephrectomy, not reimplant**.
- Stent already in place: retrograde study before removal, exchanging stent for a wire; **persistent extravasation after adequate stenting** is an indication.

> **KEY POINT** Why refluxing is acceptable in adults: VUR is rarely the indication, and refluxing anastomoses show no faster renal decline, more stenosis or (on long-term data) more infection than tunnelled ones, while needing less ureteral length and less bladder mobilisation.

Feature	Nonrefluxing (tunnelled)	Refluxing (direct)
Tunnel	**3–5 × ureteric diameter** (5:1 robotically)	None
Ureter length / bladder mobilisation	More	Less — to posterior dome
Use	High-grade reflux with renal risk	Generally sufficient in adults

Technique	Access	Neo-hiatus	Signature step
Modified **Lich-Gregoir** (nonrefluxing)	Extravesical	Posterior wall, superior-medial to native orifice	Detrusor trough over intact mucosa; detrusor closed over ureter
Modified **Politano-Leadbetter** (nonrefluxing)	Transvesical (anterior cystotomy)	New entry cephalad-medial to native hiatus	Submucosal tunnel cut with scissors; right-angle clamp draws ureter through
Refluxing extravesical	Extravesical	Posterior dome, lateralised	Direct spatulated anastomosis
Refluxing transvesical	Transvesical	Posterior dome over Kelly clamp	Clamp pulls ureter in

Open steps

- Extraperitoneal lower midline approach; find the ureter where it crosses the common iliac bifurcation (trace external iliac artery up) or beneath the obliterated umbilical artery; vessel loop.
- Divide proximal to disease (intramural dissection unnecessary); clip distal stump; clip proximal ureter to let it distend passively; 4-0 stay at 6 o'clock; frozen-section margin in stricture.
- Mobilise the bladder off the peritoneum (may divide ipsilateral medial umbilical ligament and urachus); test reach on the **decompressed** dome. Extra ureteric mobilisation buys length at the cost of devascularisation.
- **Lich-Gregoir**: partly distend bladder → small transmural cystotomy (neo-orifice) → serosa-detrusor incision toward bladder neck sparing mucosa → raise flaps (repair mucosal breach with 4-0) → spatulate anteriorly → 4-0 sutures at 12 then 6 o'clock, then interrupted → stent → close detrusor over ureter with 2-0.
- **Politano-Leadbetter**: midline anterior cystotomy → score a ≈1.5-cm mucosal defect (neo-orifice) superior-medial to native orifice → Metzenbaum scissors raise a submucosal tunnel toward the bladder neck → right-angle clamp, beak posterior, punctures the posterior wall at the tunnel's far end (entry cephalad-medial, so the ureter runs almost longitudinally) → grasp stay sutures, draw ureter through → spatulate, anastomose, stent, two-layer cystotomy closure.
- **Refluxing**: extravesical — Allis clamps on dome, transmural cystotomy in posterior dome lateralised to the pathology, spatulated anastomosis; transvesical — Kelly clamp pressed on posterior dome from inside, cystotomy over it, clamp draws ureter in. A non-transecting side-to-side refluxing anastomosis suits ureters with poor vascularity.
- Sutures: full-thickness through ureter; mucosa plus some detrusor on bladder (mucosal apposition); knots outside the lumen. Stent: **6 Fr × 26–28 cm double-J**, placed before completion.

Principles (failure = ischaemia or tension)

- Preserve periureteric adventitia (no skeletonisation); handle gently.
- Primary reimplant for defects up to **4 cm**; longer → psoas hitch/Boari; renal descent adds up to **5 cm**. More bladder reach: divide contralateral medial umbilical ligament, free peritoneum off the dome, ligate the contralateral pedicle.
- Enter **superior-medial, directed toward the bladder neck**; lateral sites kink on filling; avoid angulation at entry.
- Open surgery preferred for trauma (inspect for associated injuries) and remains the gold standard (versatile, can add hitch or flap); laparoscopic/robotic (transperitoneal, extravesical; tension tested on the non-distended dome) give less blood loss, shorter stay, equivalent patency.

Aftercare and complications

- Foley 1 week with cystogram; drain at 48 h; stent **≈6 weeks**, removed cystoscopically.
- **Leak** (non-watertight or ischaemic anastomosis): fever, pain, ileus → excretory imaging, drain urinoma → keep Foley and stent; leak after stent removal → re-stent; refractory → nephrostomy.

- **Late obstruction**, usually after stent removal (ischaemia, tension, narrow neo-orifice, poor spatulation, radiation, lateral site) → antegrade + retrograde imaging → endoscopic first for short strictures, redo surgery for high-grade.
- Others: UTI, fistula, spasm, stent colic, wound infection, bowel injury.

> **KEY POINT** Reimplantation fails for one reason in two forms — ischaemia or tension. Every rule (preserve adventitia, spatulate, enter superior-medial toward the bladder neck, bridge rather than stretch) prevents one or the other. The robot helps mainly by easing intracorporeal suturing.

7 · BRIDGING DISTAL URETERAL DEFECTS

Option	Principle	Reach	Rule
Primary reimplant	Direct	≤4 cm	Toward bladder neck
Psoas hitch	Dome fixed to psoas minor tendon	Gains ≈5 cm	Sutures superficial, parallel to fibres; reimplant before tying
Boari flap	Tubularised anterior wall flap	Up to **15 cm** (mid-ureter)	Base ≥**4 cm**, apex ≥**3 cm**, length:base ≤**3:1**, measured on empty bladder
Bladder advancement flap	Transverse incision one-third from dome to neck	Comparable to Boari	Simpler; may replace Boari
Renal descent (nephropexy)	Kidney brought down	≈5 cm	Adjunct

- Ladder: reimplant → hitch → flap. Contraindication: **small-capacity, poorly compliant bladder**. Intraoperative findings decide hitch, flap or both; hitch + flap shortens the flap and preserves capacity.
- **Hitch**: mobilise bladder (divide contralateral vas/round ligament, ± contralateral superior vesical artery, open contralateral endopelvic fascia) → oblique anterior cystotomy across about half the dome at its widest (shorter limits ascent) → finger pushes posterolateral dome onto psoas above iliac bifurcation → 4–6 2-0 sutures through tendon and detrusor avoiding **genitofemoral** (surface) and **femoral** (deep) nerves → deliver ureter, then tie → reimplant on the superolateral (relatively immobile) dome; refluxing if ureter short; 4-0 adventitial anchoring sutures, stent.
- **Boari**: oblique flap, base on diseased side, wider for longer flaps; length ≈ defect, plus ≈3 cm (3 cm overlap, 3 cm tunnel) for nonrefluxing; base anchored to psoas; refluxing anastomosis needs apex ≈3 × ureteric diameter; tubularise in two layers. Still short → more ureteric dissection or renal descent.
- Minimally invasive: reflect colon, find ureter at iliac vessels, drop bladder, hitch to psoas minor (superolateral to iliac vessels), refluxing extravesical reimplant; flap outlined with methylene-blue distension, cut starting ≈2 cm from bladder neck; 7 Fr stent.
- Specific complications: femoral/genitofemoral neurapraxia (prevented by superficial sutures along fibres), iliac vessel injury, ischaemic stricture of narrow-based flap (respect 3:1), reduced capacity with urgency/frequency/incontinence.
- Beyond reach: ureteroureterostomy, transureteroureterostomy, ileal ureter, autotransplantation.

> **KEY POINT** The bladder is the donor tissue: hitch first, flap second — every flap rule (base width, 3:1, measure empty) protects the flap's blood supply and the bladder's capacity.

8 · ENTEROCYSTOPLASTY (AUGMENTATION)

- Indication: neurogenic bladder with poor compliance **despite** clean intermittent catheterisation (CIC), anticholinergics, botulinum toxin and sacral neuromodulation — especially upper tract deterioration, hydronephrosis or incontinence with skin breakdown; rarely refractory symptoms after partial cystectomy (cystoscopy in men to exclude outlet obstruction).
- Mechanism: distensible bowel lowers pressure only if **detubularised**, which abolishes the coordinated peristalsis that would otherwise raise intravesical pressure.

- Work-up: video-urodynamics (detrusor leak point pressure predicts renal risk; shows contour, VUR, diverticula); ultrasound (hydronephrosis, atrophy); full serum chemistry, vitamin B12 and fat-soluble vitamins; renal failure worsens acidosis risk; bowel history (Crohn disease or resection → short gut; radiation → poor segment); **confirm dexterity and ability to catheterise**. Metabolic consequences → *Ch 133*.
- Concomitant: sling/artificial sphincter/bladder neck closure for stress urinary incontinence; continent catheterisable stoma (± bladder neck closure); nonrefluxing reimplant for high-grade reflux with renal dysfunction when pressure may stay high.

Segment	Why	Note
Ileum (commonest)	Abundant, mobile, long mesentery, robust supply	25 cm (15–20 cm minimally invasive) taken **≈20 cm** proximal to ileocaecal valve (spares terminal ileum: B12/bile salts); acidosis risk
Ileocaecal	When a catheterisable channel is wanted (e.g. after bladder neck closure)	—
Sigmoid	Redundant mesentery; neurogenic patients	Colonoscopy first
Stomach	Possible	Higher incision

ALGORITHM
Midline incision, peritoneum opened → isolate ileum, check tension-free reach to dome → transilluminate mesentery (arcade reaches ends) → staple bowel; divide mesentery keeping a broad base (no torsion) → side-to-side stapled anastomosis; close mesenteric window (internal hernia) → excise staple lines, irrigate, **detubularise** on antimesenteric border, fold into U → **bivalve bladder widely** along the dome, anterior to posterior → wide-mouthed transmural anastomosis (robotically: start posteriorly, U apex at bladder neck) → suprapubic tube through separate cystotomy ± Foley → leak test, drain.

- A narrow mouth forms a poorly emptying pseudodiverticulum.
- Early: extravasation (inadequate suturing or flap ischaemia — preserve the arcade), UTI, bowel injury, fistula, intra-abdominal infection (isolate field when opening bowel).
- **Perforation**: missed catheterisation; pain, fever, distension, often masked by sensory loss → CT cystogram → **emergency exploration**. Stresses lifelong regular CIC.
- Stones (mucus, bacteria, stasis) — commonest long-term complication; irrigation may reduce mucus. Neoplasia (adenocarcinoma and urothelial) → surveillance endoscopy from **10 years**. Routine blood tests for metabolic and vitamin deficiency.
- Cannot catheterise at the bladder neck (habitus) → continent ileocaecocystoplasty; unable to catheterise → incontinent ileovesicostomy.
- Long-term: high continence and satisfaction, but secondary procedures often needed.

KEY POINT Augmentation exchanges a pressure problem for a lifelong bowel-in-bladder problem: detubularisation and a wide-mouthed anastomosis win the pressure battle; catheterisation, irrigation, blood tests and endoscopic surveillance manage the price.

9 · CONTINENT CATHETERISABLE CHANNELS

- For bladders of adequate capacity needing chronic catheterisation (neuropathic, refractory urethral stricture), usually after failed urethral CIC; combine with augmentation if compliance or capacity poor. Assess dexterity, habitus, neurological condition, bowel history (inflammatory bowel disease, radiation, prior appendicectomy or resection), renal function, video-urodynamics.
- **Continence lives in the submucosal detrusor tunnel**, not the conduit. Channel must reach the stoma (right lower quadrant or umbilicus) without tension, angulation or loss of blood supply. Always have a fallback plan (appendix absent or short).

- Vas, colon and stomach channels: difficult or poorer outcomes.

Feature	Appendicovesicostomy (Mitrofanoff, 1980)	Reconfigured ileum (Yang-Monti)
Status	First choice if present and long enough	Appendix absent, short or reserved (e.g. for antegrade continence enema)
Conduit	Natural tube, ideal calibre, no metabolic effect	2–4 cm ileum detubularised, retubularised **transversely** over a catheter (longer, narrower tube)
Extra length	Caecal cuff (also widens stoma); mobilise caecum and ascending colon	Double Monti (channel diverticula, difficult catheterisation) or spiral Monti (more revisions)
Minimally invasive	Widely reported, mainly children (≈4 cm tunnel, 4-0 polydioxanone anastomosis)	Rare — demanding

ALGORITHM
Minimally invasive appendicovesicostomy: lithotomy, 10–15° Trendelenburg → stay suture on appendix → detach from caecum with cuff, close caecal defect → spatulate distal end → ≈4-cm submucosal detrusor tunnel at dome or anterior wall → button-hole mucosa, anastomose with 4-0 polydioxanone → catheter in channel, stoma matured in right lower quadrant or umbilicus, pelvic drain.

Open: lower midline or Pfannenstiel; appendix detached with a caecal cuff (defect closed), distal end spatulated and tunnelled into the dome or anterior wall, intra- or extravesically. Catheter left in channel; bladder drained (Foley or suprapubic) **2–4 weeks** before channel catheterisation starts. Continence **>95%** for both; commonest complication **stomal stenosis**.

> **KEY POINT** Use the appendix when present and long enough; reconfigure ileum only when it is not. Either way, continence comes from tunnel length through detrusor, and failure to catheterise regularly risks the bladder, not just the channel.

10 · VESICOVAGINAL FISTULA (VVF) †

All facts in this section are outside the source notes (†).

- **Causes:** worldwide, obstructed labour (ischaemic necrosis). In developed countries, **hysterectomy** is the commonest cause (fistula at the vaginal vault, supratrigonal); also radiation, pelvic malignancy and caesarean section.
- **Presentation:** continuous painless leakage per vagina, usually 1–3 weeks after hysterectomy; radiation fistulae can appear years later.
- **Diagnosis:** speculum examination and cystoscopy (site, size, distance from the ureteric orifices). **Double-dye tampon test**: oral phenazopyridine plus intravesical methylene blue — blue staining = VVF; orange staining of the upper tampon = ureterovaginal fistula. Image the upper tracts (CT urography): a coexisting ureteric injury or fistula is present in about 10%. Biopsy any fistula after radiation or cancer.
- **Conservative:** a small, fresh, non-irradiated fistula may close with 2–6 weeks of continuous catheter drainage.
- **Timing of repair:** early repair is acceptable if the tissue is healthy and uninfected. Otherwise wait about 3 months; for radiation fistulae wait 6–12 months and plan a vascularised flap.
- **Vaginal route** (most post-hysterectomy fistulae): flap-splitting multilayer repair or Latzko partial colpocleisis; interpose a Martius labial fat pad when tissue is poor.
- **Abdominal route** (transvesical O'Conor bivalve, open or robotic): high or large fistulae, need for ureteric reimplantation or augmentation, or failed vaginal repair; interpose omentum.
- **Principles of any repair:** good exposure, excise or freshen the tract, tension-free watertight closure in layers, non-overlapping suture lines, a vascularised interposition flap when needed, and continuous bladder drainage for 10–14 days with a cystogram before catheter removal.

- **Ureterovaginal fistula:** try a ureteric stent first; if it fails, ureteroneocystostomy (± psoas hitch).

11 · MCQ ONE-LINERS

1. Bladder diverticulum wall lacks functional muscularis propria.
2. Hutch diverticulum: congenital, posterolateral, beside ureteric orifice, with VUR.
3. Cancer within a diverticulum → partial or radical cystectomy, not diverticulectomy.
4. Endoscopic prostate surgery precedes diverticulectomy at the same sitting.
5. Cleavage plane in diverticulectomy: sac–pseudocapsule plane.
6. Combined approach hazard: incisions not meeting → excess bladder wall excised.
7. Commonest complication of diverticulectomy and partial cystectomy: urinary extravasation.
8. Urine leak confirmed by drain-fluid creatinine exceeding serum creatinine.
9. Best imaging for bladder stone size: non-contrast CT.
10. Percutaneous cystolithotripsy avoided after prior abdominal surgery or augmentation (bowel injury).
11. Prolonged transurethral cystolitholapaxy risks urethral stricture.
12. Percutaneous cystolithotripsy sheath: 30 Fr; stones <1 cm removed intact.
13. Bladder paraganglioma: syncope on micturition; α-block before instrumentation.
14. Paraganglioma cystoscopy: yellow raised mucosa; MIBG confirms function.
15. Bladder endometriosis is mainly extravesical; treated by partial cystectomy.
16. Urachus = median umbilical ligament; excise with dome cuff, no wide margin.
17. IC/BPS cystectomy success: small fibrotic bladder, Hunner lesions.
18. Simple cystectomy preserves prostate; no lymphadenectomy.
19. Poorly functioning kidney with distal stricture → nephrectomy, not reimplant.
20. Nonrefluxing tunnel length: 3–5 times ureteric diameter.
21. Neo-hiatus superior-medial, toward bladder neck; lateral sites kink.
22. Reimplant stent: 6 Fr double-J for about 6 weeks.
23. Primary reimplant ≤4 cm; psoas hitch gains ≈5 cm; Boari up to 15 cm.
24. Psoas hitch sutures parallel to fibres: spare genitofemoral and femoral nerves.
25. Reimplant the ureter before tying psoas hitch sutures.
26. Boari flap: base ≥4 cm, length:base ≤3:1, measured empty.
27. Psoas hitch/Boari contraindicated with small, non-compliant bladder.
28. Ileal augment: take segment ≈20 cm proximal to ileocaecal valve.
29. Detubularisation abolishes peristalsis; narrow anastomosis forms pseudodiverticulum.
30. Augment perforation: missed catheterisation; emergency exploration.
31. Augmented bladder: surveillance cystoscopy from 10 years.
32. Channel continence depends on submucosal detrusor tunnel.
33. Yang-Monti: 2–4 cm ileum retubularised transversely.
34. Commonest channel complication: stomal stenosis.

★ Putting it together: the big treatment choices

Reading time ≈ 8 min (slow pace) · Integrates Ch 124, 129–135 · use these tables as the skeleton of long-essay answers

A · NON-MUSCLE-INVASIVE BLADDER CANCER (NMIBC): RISK → TREATMENT

After complete transurethral resection of bladder tumour (TURBT), the American Urological Association / Society of Urologic Oncology (AUA/SUO) risk group decides adjuvant therapy and surveillance. Low grade (LG) and high grade (HG) refer to the 2004 World Health Organization grading.

Risk	Defined by	Adjuvant treatment	Cystoscopy
Low	LG solitary Ta ≤3 cm; papillary urothelial neoplasm of low malignant potential (PUNLMP)	Single post-op chemotherapy instillation	3 months, then yearly from 12 months
Intermediate	LG Ta recurring within 1 yr, >3 cm or multifocal; LG T1; HG Ta ≤3 cm	6-week induction chemotherapy (mitomycin C or gemcitabine) or bacillus Calmette–Guérin (BCG); maintenance to 1 yr	3–6-monthly for 1–2 yr, then less often
High	HG T1; HG Ta >3 cm, multifocal or recurrent; any carcinoma in situ (CIS); variant histology; lymphovascular invasion; HG prostatic urethra	Re-TURBT for T1; induction BCG + **maintenance to 3 yr** (SWOG schedule); early radical cystectomy (RC) for the highest risk	3–4-monthly for 2 yr → 6-monthly for 2 yr → yearly for life; upper-tract imaging yearly for 2 yr
BCG-unresponsive	Persistent/recurrent HG disease despite adequate BCG (≥5/6 induction + ≥2/3 maintenance)	RC is standard; if unfit or refusing: pembrolizumab (CIS), nadofaragene, gemcitabine–docetaxel	—

B · MUSCLE-INVASIVE BLADDER CANCER (MIBC): CYSTECTOMY VS BLADDER PRESERVATION

	Neoadjuvant chemotherapy (NAC) + RC + pelvic lymph node dissection (PLND)	Trimodality therapy (TMT)
Who	Default for fit patients. NAC (cisplatin-based: gemcitabine–cisplatin or dose-dense methotrexate, vinblastine, doxorubicin, cisplatin [ddMVAC]) only if cisplatin-eligible	Solitary T2 tumour, visibly complete TURBT, **no CIS, no hydronephrosis**, good bladder function; or unfit for RC
Cisplatin-ineligible if any	GFR <60, Eastern Cooperative Oncology Group (ECOG) performance status ≥2, grade ≥2 hearing loss or neuropathy, New York Heart Association (NYHA) class III–IV heart failure → straight to local therapy	
After treatment	Adjuvant nivolumab (CheckMate 274) for ≥pT3/N+ after NAC or if no NAC given	Lifelong cystoscopy; salvage RC in 10–30%
See	Ch 131, 132	Ch 131

C · CHOOSING THE URINARY DIVERSION

	Ileal conduit	Orthotopic neobladder	Continent cutaneous reservoir
Best for	Frail, poor renal function, unwilling or unable to self-catheterise, neurological disease	Default continent option in fit patients of either sex	Urethra unusable (tumour at urethral margin, stricture) or incompetent sphincter

	Ileal conduit	Orthotopic neobladder	Continent cutaneous reservoir
Must have	Usable ileum (not irradiated or diseased)	eGFR **>35–40**; intact sphincter; negative urethral frozen section; willing to self-catheterise	eGFR >35–40; dexterity; reliable efferent limb (appendix, tapered ileum)
Main problems	Stoma complications, uretero-ileal stricture	Night incontinence; retention (women, late); acidosis	Stomal stenosis, catheterisation difficulty, stones
See	Ch 133, 135	Ch 134, 135	Ch 134

D · METABOLIC COMPLICATIONS BY BOWEL SEGMENT

Segment	Electrolyte picture	Treatment
Ileum / colon	**Hyperchloraemic metabolic acidosis** (ammonium and chloride absorbed)	Alkali (sodium bicarbonate, potassium citrate); chloride-transport blocker (chlorpromazine, nicotinic acid)
Jejunum	**Hyponatraemic, hypochloraemic, hyperkalaemic acidosis**	Sodium chloride, thiazide; avoid jejunum
Stomach	**Hypochloraemic, hypokalaemic metabolic alkalosis**; haematuria–dysuria syndrome	H2 blocker / proton-pump inhibitor

Loss of terminal ileum also causes vitamin B12 deficiency and bile-salt diarrhoea. Chronic acidosis buffers through bone, which leads to osteomalacia. → *Ch 133*

E · METASTATIC UROTHELIAL CARCINOMA IN FOUR STEPS

ALGORITHM

1. Fitness: cisplatin-eligible, cisplatin-ineligible but platinum-eligible, or platinum-ineligible.

2. First line: enfortumab vedotin (EV) + pembrolizumab (EV-302); otherwise gemcitabine–cisplatin (or gemcitabine–carboplatin if cisplatin-ineligible).

3. After platinum response or stable disease: avelumab maintenance (JAVELIN Bladder 100).

4. Later lines: pembrolizumab; erdafitinib if FGFR-altered; EV; sacituzumab govitecan. Platinum-ineligible: pembrolizumab first. → *Ch 131*